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Integrating evidence, models and maps to enhance Chagas disease vector surveillance

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Abstract:	Background Until recently, the Chagas disease vector, Triatoma infestans, was widespread in Arequipa, Peru. As a result of a decades-long campaign in which over 70,000 houses were treated with insecticides, infestation prevalence is now greatly reduced. To monitor for T. infestans resurgence, the city is currently in a surveillance phase in which a sample of houses is selected for inspection each year. Despite extensive data from the control campaign that could be used to inform surveillance, the selection of houses to inspect is often carried out haphazardly or by convenience. Therefore, we asked, how can we enhance efforts toward preventing T. infestans resurgence by creating the opportunity for vector surveillance to be informed by data?

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	Methodology/Principal Findings To this end, we developed a mobile app that provides vector infestation risk maps generated with data from the control campaign run in a predictive model. The app is intended to enhance vector surveillance activities by giving inspectors the opportunity to incorporate the infestation risk information into their surveillance activities, but it does not dictate which houses to surveil. Therefore, a critical question becomes, will inspectors use the risk information? To answer this question, we ran a pilot study in which we compared surveillance using the app to the current practice (paper maps). We hypothesized that inspectors would use the risk information provided by the app, as measured by the frequency of higher risk houses visited, and qualitative analyses of inspector movement patterns in the field. We also compared the efficiency of both mediums to identify factors that might discourage risk information use. Over the course of ten days (five with each medium), 1,081 houses were visited using the paper maps, of which 366 (34%) were inspected, while 1,038 houses were visited using the app, with 401 (39%) inspected. Five out of eight inspectors (62.5%) visited more higher risk houses when using the app (Fisher's exact test; $p < 0.001$). Among all inspectors, there was an upward shift in proportional visits to higher risk houses when using the app (Mantel-Haenszel test, common odds ratio (OR) = 2.42, 95% CI 2.00-2.92), and in a second analysis using generalized linear mixed models, app use increased the odds of visiting a higher risk house 2.73-fold (95% CI 2.24-3.32), suggesting that the risk information provided by the app was used by most inspectors. Qualitative analyses of inspector movement revealed indications of risk information use in seven out of eight (87.5%) inspectors. There was no difference between the app and paper maps in the number of houses visited (paired t-test, $p = 0.67$) or inspected ($p = 0.17$), suggesting that app use did not reduce surveillance efficiency. Conclusions/Significance Without staying vigilant to remaining and re-emerging vector foci following a vector control campaign, disease transmission eventually returns and progress achieved is reversed. Our results suggest that, when provided the opportunity, most inspectors will use risk information to direct their surveillance activities, at least over the short term. The study is an initial, but key, step toward evidence-based vector surveillance.
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Competing Interests	The authors have declared that no competing interests exist.

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September 19th, 2018

Dear Editors,

Thank you for the opportunity to revise our manuscript, ""Integrating evidence, models and maps to enhance Chagas disease vector surveillance" to be considered for publication as a Research Article in the PloS NTDs Vectors collection. The reviewers' comments are appreciated, and have helped us improve our manuscript. We have included the point-by-point responses to each reviewer comment in the document entitled, 'Response to reviewers.'

Sincerely,
Jennifer K. Peterson, PhD, on behalf of all authors

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Integrating evidence, models and maps to enhance Chagas disease vector surveillance

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Keywords

Entomological surveillance, spatial epidemiology, vector-borne disease, triatomine bugs, *Triatoma infestans,* Chagas disease, mHealth, risk maps, implementation science

Abstract

Background

Until recently, the Chagas disease vector, *Triatoma infestans*, was widespread in Arequipa, Peru, but as a result of a decades-long campaign in which over 70,000 houses were treated with insecticides, infestation prevalence is now greatly reduced. To monitor for *T. infestans* resurgence, the city is currently in a surveillance phase in which a sample of houses is selected for inspection each year. Despite extensive data from the control campaign that could be used to inform surveillance, the selection of houses to inspect is often carried out haphazardly or by convenience. Therefore, we asked, how can we enhance efforts toward preventing *T. infestans* resurgence by creating the opportunity for vector surveillance to be informed by data?

Methodology/Principal Findings

To this end, we developed a mobile app that provides vector infestation risk maps generated with data from the control campaign run in a predictive model. The app is intended to enhance vector surveillance activities by giving inspectors the opportunity to incorporate the infestation risk information into their surveillance activities, but it does not dictate which houses to surveil. Therefore, a critical question becomes, will inspectors use the risk information? To answer this question, we ran a pilot study in which we compared surveillance using the app to the current practice (paper maps). We hypothesized that inspectors would use the risk information provided by the app, as measured by the frequency of higher risk houses visited, and qualitative analyses of inspector movement patterns in the field. We also compared the efficiency of both mediums to identify factors that might discourage risk information use.

Over the course of ten days (five with each medium), 1,081 houses were visited using the paper maps, of which 366 (34%) were inspected, while 1,038 houses were visited using the app, with 401

(39%) inspected. Five out of eight inspectors (62.5%) visited more higher risk houses when using the app (Fisher's exact test; $p < 0.001$). Among all inspectors, there was an upward shift in proportional visits to higher risk houses when using the app (Mantel-Haenszel test, common odds ratio (OR) = 2.42, 95% CI 2.00-2.92), and in a second analysis using generalized linear mixed models, app use increased the odds of visiting a higher risk house 2.73-fold (95% CI 2.24-3.32), suggesting that the risk information provided by the app was used by most inspectors. Qualitative analyses of inspector movement revealed indications of risk information use in seven out of eight (87.5%) inspectors. There was no difference between the app and paper maps in the number of houses visited (paired t-test, $p = 0.67$) or inspected ($p = 0.17$), suggesting that app use did not reduce surveillance efficiency.

Conclusions/Significance

Without staying vigilant to remaining and re-emerging vector foci following a vector control campaign, disease transmission eventually returns and progress achieved is reversed. Our results suggest that, when provided the opportunity, most inspectors will use risk information to direct their surveillance activities, at least over the short term. The study is an initial, but key, step toward evidence-based vector surveillance.

Author Summary

Chagas disease is a serious infection that is spread by blood-sucking insects called 'kissing bugs.' These bugs live in and around human homes, and until recently, they infested thousands of human homes throughout Arequipa, the second largest city in Peru. However, a decades-long control campaign drastically reduced the number of infested houses, and the city is now in a stage where health personnel annually inspect a sample of houses throughout the city for kissing bug

reinfestation. A large amount of information was collected during the control campaign that could be used to help identify the houses at highest risk for re-infestation, so we developed a cell phone app to provide this information to health personnel in the form of interactive, user-friendly risk maps. We carried out a pilot study to see if health personnel would use these maps to select houses to inspect for re-infestation, and we found that most inspectors did use the information. We also observed that using the app did not slow the inspectors down, which can be an issue when introducing new technology. Our results suggest that the app could be a useful tool for monitoring diseases spread by insects in cities.

Introduction

Background

Chagas disease is a neglected tropical disease (NTD) endemic to the Americas with a current estimated prevalence of six to nine million people worldwide [1,2] and 70 million more at risk [3]. An estimated 30% of those with Chagas disease will develop serious cardiac and/or gastrointestinal problems for which there is no vaccine or cure [4,5] The etiological agent of Chagas disease, *Trypanosoma cruzi,* is a parasite of mammals that is transmitted between vertebrate hosts by triatomine bugs [6], and vector control is at the core of large-scale Chagas disease control efforts [7–9].

Historically, Chagas disease was considered to be a rural problem [5,10,11] associated with homes made of rudimentary materials [12–15], the presence of domestic animals in and around the domicile [16,17], and/or close proximity to less disturbed landscapes that serve as habitat for sylvatic mammal reservoirs of *T. cruzi* and *T. cruzi* vector foci [18,19]. Disease control efforts in the past were designed accordingly, to accommodate the characteristics of rural areas. However,

Chagas disease is now known to be established in several urban settings, creating a new epidemiological challenge for prevention [20–28].

In Arequipa, Peru, with a population approaching one million people, Chagas disease is an urban problem due to widespread domestic infestation by the triatomine bug species *Triatoma infestans* [23,28–37]. In 2002, a vector control campaign targeting *T. infestans* was implemented in Arequipa, and today the bug is nearly eliminated from the city. The campaign is now in its 16th year; over 70,000 households in 16 out of 18 target districts have been treated with insecticides in what was called the 'attack' phase of the campaign. These houses are now in the surveillance phase of the campaign, in which the highly challenging task of monitoring for vector resurgence is carried out through annual inspections of a fluctuating proportion of houses in each district. Although the 'attack' phase of the campaign generated a great amount of data relevant to the risk of subsequent vector infestation [30], these data are rarely used to inform the selection of houses to visit in the surveillance phase. Rather, the selection of houses is often carried out haphazardly or by convenience. Therefore, we asked, how can we harness the extensive data collected during the attack phase to enhance vector surveillance, and continue the considerable progress made toward the elimination of *T. infestans* from Arequipa?

VectorPoint: infestation risk maps to support independent decision making

To this end, we developed a cloud-based, open-source mobile app that provides vector infestation risk maps for use by health inspectors. The app, which we call 'VectorPoint,' is intended to enhance vector surveillance by giving inspectors the opportunity to incorporate infestation risk information into their process of selecting houses to inspect for *T. infestans*. Risk information is generated by a predictive model that calculates infestation risk estimates using data from the attack phase of the control campaign, in combination with new data collected during the surveillance phase. The app

also provides a data entry function to collect new surveillance data. Upon collection, new data are sent directly to a virtual server and then incorporated into the next run of the model, after which they are immediately visualized in the risk maps.

There are currently several apps for disease surveillance in resource limited settings, the most common being SMS-based apps (FrontlineSMS [38–40], RapidSMS [41,42], U-Report [43,44], Ushahidi [40,45], CycleTel [46,47], Geochat [48], among others [49]; see [50] for a thorough review of SMS apps for disease surveillance), and generic software and tool collections that offer mobile device-based data collection as their primary function, and some combination of basic data analysis, visualization and/or mapping as secondary functions (SAGES [51] Open data kit [52–54]; Epicollect [54–56]; eMOCHA [57,58]; Medic mobile [59]; Magpi (formerly Episurveyor) [60–62]; DataWinners [63]; and PhiCollect [64], among others [65–69]). A small number of apps have been developed for vector surveillance (Dengue Chat [70]; CHAAK [71], eMOCHA [72]), with the primary features being data collection based in social networking and community based surveillance [70] and data collection using electronic forms and/or SMS [71,72].

VectorPoint is unique in that it supports independent decision making by the individual collecting the data, and it does not dictate a path to the user or mandate which houses to surveil. Rather, it provides the opportunity to integrate risk information into the inspector's decision making process. Collaborative approaches that give control to the end-user have been shown to contribute to the sustainability of new technologies in resource-limited settings [73], and this is an important feature of VectorPoint. However, an inherent challenge with implementing a technology that supports independent front-end user decision-making is that the user can decide not to use it. As such, the potential for the app to enhance vector surveillance lies in the hands of the front-end user, and a critical question becomes, will inspectors use the information it provides?

VectorPoint Pilot

To answer this question, we carried out a pilot trial comparing surveillance using the VectorPoint app to the current practice of surveillance using paper maps. We hypothesized that inspectors would use the risk information provided by the app, as measured by the frequency of higher risk houses visited and inspected with the app compared to the paper maps. We also looked for qualitative evidence of risk information use by analyzing daily and weekly maps of the inspector's movement patterns throughout the search zones when using the app and paper maps. Finally, we compared measures of productivity between the app and the paper maps to ensure that the app was not hindering inspector progress, which might also discourage its use.

Methods

App Overview

The front end (i.e., what the user sees and interacts with) of VectorPoint is a neighborhood map that displays *T. infestans* infestation risk at the individual house level, and a data entry tool for collecting data resulting from home inspections (Figure 1). The back end of VectorPoint (Figure 2) is composed of a spatio-temporal Gaussian field model that generates the infestation risk estimates visualized in the maps and a relational system of cloud-based databases and servers that are used to store and send data between the predictive model and the platform that visualizes the model-generated risk estimates in the maps. Below is a more detailed description of each component of VectorPoint.

Figure 1. *T. infestans* infestation risk map zoomed out (left) and zoomed in (right). Houses are represented by dots that are colored according to their risk of infestation as estimated by the model. Legend translates to (from top to bottom): 'Risk of infestation- Lowest; Low;

Medium; High; Highest.' Note that images show infestation data overlay and app functions only; cartographic details (i.e., roads, parks, etc) have been removed for this publication. Images of the app with cartographic detail are available in the VectorPoint repository: https://github.com/chirimacha/VectorPoint.

Infestation risk map

The principal feature of VectorPoint is the risk map (Figure 1), which displays *T. infestans* relative risk estimates at the household level that are generated by a statistical model (detailed below). The map is intended to be used by health inspectors carrying out house-to-house *T. infestans* surveillance. The output from the model is presented in a simple and user-friendly format in which risk estimates are divided into 5 quantiles, ranging from lowest to highest infestation risk, and each quantile is then represented in the map by a color. We use the 5-class, multi-hue color scheme, 'YlOrRd,' developed by Brewer [74] for cartography data visualization [75,76], which is color-blind friendly. The color scheme progresses from light yellow to dark red, (a color progression associated with increasing hazard among Spanish speakers [77]) with saturation increasing with infestation risk, and orange representing intermediate risk. Each house is displayed in the map as a dot that is colored as one of the five colors that corresponds to the its infestation risk estimate. A legend in the corner of the map presents the colors accompanied by a one or two word description in Spanish of their corresponding infestation risk category, which translate into English as, "lowest," "low," "medium," "high," and "highest" infestation risk. Each risk category is represented in the map equally. The maps are set to display relative risk (i.e., a house's risk of infestation relative to all other houses in the neighborhood), but they can be adjusted from the back end to display data divided into any number of quantiles, or to display absolute, instead of relative, risk estimates.

Data collection tool

The second feature of VectorPoint is its data collection functionality (Figure 2). Inspectors can enter the data resulting from individual home visits and inspections directly into a data entry form in the app. The form is designed to collect the same data as the paper forms used by the Ministry of Health for *T. infestans* surveillance: date, house code (explanation below), areas of the home inspected (inside, outside, or both), number of inhabitants, number and type of domestic animals, and whether *T. infestans* or signs thereof (generally, eggs, feces or exuviae, grouped together as 'rastros,' meaning 'traces') were found. Radio buttons and drop-down menus are provided whenever possible for consistency, and to avoid typographical errors. After the data entry form is completed, data are encrypted and transmitted from the app to a SQL database, eliminating the step of digitizing data from paper forms.

Figure 2. Part of the data collection form found in VectorPoint. Spanish text translates to, from top to bottom: 'Date'; 'Property characteristics: Regular house'; 'State of the inspection: inspection'; 'Inspection area: *Triatoma infestans* ('chiris'), signs of the bugs ('rastros'), inside the house ('intra'), yard and patio ('peri')'; 'How many people live on the property?'; 'What animals are there? : Dogs, Cats, Poultry, Guinea Pigs, Rabbits.'

It should be noted that *T. infestans* surveillance data in Arequipa are organized with 4 tuple identification (ID) codes assigned to each home by the Peruvian Ministry of Health at the beginning of the vector-control campaign. The 4 tuple consists of: province/district/locality/house. (VectorPoint is designed to be used for house to house surveillance at the locality level, which are neighborhoods ranging from 30-2000 households.) We have maintained the 4 tuple ID system in VectorPoint, and throughout the manuscript, we refer to the 4 tuple house IDs as 'house codes.'

Spatio-temporal model

The model in VectorPoint is designed to estimate the relative probability of *T. infestans* infestation of sites (primarily households) in an urban landscape. The model incorporates three types of information: (i) site covariates; (ii) the results of any previous inspections for *T. infestans*; and (iii) infestation history in neighboring sites. For each site, we include one covariate that is an indicator of participation in the attack phase of the vector control campaign, during which insecticide was applied to all participating households, as previous studies have shown that houses that did not participate are more likely to be infested [30]. We did not include other finer-scale risk factors for *T. infestans* infestation, such as guinea pig husbandry [23], because data were not available at the scale required for app.

Concretely, let the probability of vector presence, *i,* at time, *t*, be given by π_i. We model the probability using a logistic model with intercept, β_0, covariate information, β_1, and separable spatio-temporal random effects, u_{it} and v_{it}:

$$logit(\pi_i) = \beta_0 + \beta_1 + u_{it} + v_{it}$$

where u_{it} is a realization of the Gaussian field with a Matérn covariance structure [78,79]. The Gaussian field functions such that any adjustment to the estimate for one house affects all other houses in a given area, with a greater effect on those nearby. The Matérn function is a versatile model of covariance that includes Gaussian covariance as a special case [80]. The term v_{it} is a first order autoregressive discrete time random effect.

As mentioned earlier, the model takes into account the inspection history of each household/site. We currently include four discrete time periods (Figure 3). We selected our time periods in reference to the phase of the vector control campaign in each area. The earliest point reflects the 'attack' phase of the campaign, which occurred between January 11th, 1997 and January 6th, 2014,

257 depending on district. The second time period is the early surveillance period, and it includes all
258 inspection data collected between January 7th, 2014 and January 6th, 2016. The third time period
259 includes inspection data collected between January 7th, 2016 and January 4th, 2018. The final time
260 point reflects the current calendar year, currently set at January 5th, 2018 - present. The later time
261 periods can be adjusted if needed. The predicted probability of infestation for most recent time
262 point is visualized in the app.
263
264 **Figure 3. Map of a locality as represented in the model throughout the four time periods (T1-**
265 **T4). Dots represent houses. T1 represents the time period of the 'attack phase' of the control**
266 **campaign, which occurred between January 11th, 1997 and January 6th, 2014, depending on**
267 **district. In T1, grey dots represent houses that participated in the 'attack phase' of the**
268 **control campaign but were not found to be infested with *Triatoma infestans*, and black dots**
269 **represented houses that participated and were found to be infested with *T. infestans.* Time**
270 **periods T2-T4 represent the time period of the surveillance phase (T2: January 7th, 2014 -**
271 **January 6th, 2016; T3: January 7th, 2016 - January 4th, 2018; T4: the current calendar year,**
272 **currently set at January 5th, 2018 - present). In T2-4, grey dots represent houses that were**
273 **inspected and not found to be infested with *T. infestans* during the surveillance phase.**
274
275 We fit the model with integrated nested Laplace approximations (INLA) using the R-package,
276 "INLA" [81,82]. To account for effects of streets as semi-permeable barriers to the spread of *T.*
277 *infestans* [29], we used an extension of a Gaussian Field model in which we stretched the city map
278 so the geographic center, (i.e., the statistical mean of the coordinates), of each city block is at a
279 multiple (1.5;[83]) of the true distance. We maintain the within-block structure, so only the
280 distance between blocks is stretched [83]. We set strong priors (mean =1.17 and standard
281 deviation= 0.01) on the covariate of not participating in the original insecticide application
282 campaign, based on our previous analysis of this factor [30]. We set the prior on the intercept term

to correspond to an expected baseline infestation prevalence of approximately 1 in 1000, with the precision matrix set to 50. This value reflects our best estimate of *T. infestans* infestation prevalence in Arequipa based on recent results from both passive surveillance (i.e., reports of *T. infestans* infestation from community members that are later confirmed by health personnel) and active surveillance (house to house surveillance conducted by our team and the Ministry of Health).

Data flow and platform

Infestation risk estimates generated by the model are sent to a cloud-based database (Amazon Relational Database Service from Amazon Web Services) through the RMySQL package [84] . These data are then sent to the Shiny [85] server, which graphically renders the risk estimates in the app. Inversely, new data collected with the app are sent back to the SQL database, and incorporated into the next run of the model. We present a diagram of the VectorPoint workflow in Figure 4.

Figure 4. VectorPoint workflow diagram. From left, risk estimates generated by the model are first sent to an SQL database, and then sent to the Shiny platform and server for visualization in the risk map. From right, data collected by the app are sent to the Shiny server and then to the SQL database. The model then pulls the new data from the SQL database the next time it is run. All data are TLS encrypted.

To address connectivity issues, VectorPoint has a caching function that stores partial inspection results, and retrieves them when connectivity is restored. VectorPoint supports multiple model result tables, and all operations are computationally parallel, allowing the app to be used by multiple inspectors simultaneously without speed degradation.

We constructed VectorPoint using open source software throughout to enable sharing and extensions. We built it using the Shiny package for R [85], and we implemented it entirely in the

open source R programming language [82]. We mapped vector infestation predictions using the leaflet package for R [86], and we overlaid these data on top of street data from OpenStreetMap.org. Open source code and related tools for VectorPoint can be downloaded from: https://github.com/chirimacha/VectorPoint, including a link to a fully-functional installation of the app, which is available in the 'README' section.

Workflow in the field

VectorPoint is located on a web server that can be accessed using a web browser on any desktop or mobile device, regardless of platform (Android, iOS, Windows, OSX, Linux,etc). Upon loading the VectorPoint web page, the inspector is presented with an authentication form in which they enter a username and password. All connections are encrypted, and risk maps can be accessed only by the study team and authorized health personnel. After user authentication, the inspector selects the locality or group of localities where they will carry out surveillance that day. The app retrieves the data for the locality from the database, and loads the corresponding risk map, zoomed out (Figure 1a). This view provides the inspector with a high-level view of the houses and their relative risk levels for infestation.

From there, the inspector zooms in on the map (Figure 1b), and selects a house to potentially visit by clicking on the corresponding dot on the map. A dialogue box will open up containing the house code, the date that the house was last visited, and whether or not the house was inspected at that time. If the inspector decides to visit that house, they can load a data entry form with the house code and date auto-filled in. If the inspector receives permission to inspect the house for *T. infestans,* data from the inspection is entered into the data entry form. If the inspector does not receive permission to inspect the house, it is recorded in the data entry form as one of four alternative outcomes: "interview," "closed," "refused," or "return." 'Interview' means that the inspector spoke with someone at the door about *T. infestans* infestation, but did not receive

permission to enter the house and inspect it; 'closed' means that no one answered the door; 'refused' means that inspection was directly refused; and 'return' means that the inspector was asked to return at a later time. After each house visit, data are sent from the app to the database, regardless of visit outcome. In cases of data outages or other internet connectivity issues, inspection data can be saved and sent to the database at a later time, as described above. This process is repeated for each home visited by the inspector in a given day. At the end of the day, all data collected with the app are pulled from the database and run in the model to generate new predictions. The predictions are then pushed back to the database and visualized in the map.

Field study design

In the study comparing *T. infestans* surveillance with the app to surveillance under the current practice of using hand-drawn paper maps (Figure 5), eight members of our field team previously trained to carry out home inspections for *T. infestans* carried out vector surveillance in Arequipa for a total of two work weeks (10 days). At the beginning of each week, inspectors were randomly assigned (i) a zone to surveil, and (ii) if they would use the app or the paper map in that zone. Only one inspector was assigned to each search zone, which were all in the same city district. Each search zone met the following five criteria: (i) it was located in a developed area (i.e., all roads paved) within the central portion of the district, (which is safer than peripheral, less developed areas); (ii) it contained a minimum of 400 houses, and no more than 1.25 times the number of houses in the zone with the fewest houses; (iii) area was a minimum of 0.1 km^2, and could not be greater than twice that of the zone with the smallest area; (iv) house density was at least 2000 houses per km^2; and (v) the search zone was in a locality where at least one house had been found positive for *T. infestans* during the attack phase of the control campaign. These criteria resulted in 16 search zones with 416-514 homes, an area of 0.12-0.20 km^2, a perimeter of 1.62-2.37 km, and house density ranging from 2,570 to 3,623 houses per km^2. Half of the inspectors used the app in the first week and paper maps in the second week, and the other half used paper maps in the first

week and the app in the second week. All inspectors used the same cell phone model and operating system when using the app (Samsung Galaxy J7, with Android version 7.0), to control for variation between devices. Inspectors received training in app use with these phones before starting the experiment.

Figure 5. Example of a paper map used in vector surveillance under the current practice. The numbers displayed on each block are the last three digits of the house code, with arrows indicate their ascending order. For example, a block with '1 → 12' indicates that houses with unicodes one through 12 are on that block. Street names and other identifying information have been removed.

Surveillance was carried out daily during normal working days (Monday-Friday) and hours (7am-1pm). Inspectors were told to carry out *T. infestans* surveillance as they would normally, and that their objective was simply to find *T. infestans,* in order to avoid perceptions that they needed to fulfill a quota of visiting a certain number of houses. The protocol for *T. Infestans* inspections is stipulated by the Peruvian Ministry of Health, and consists of systematic searches in all areas of the domicile and peri-domicile (pending permission by the resident), including spaces occupied by humans and live animals. Searches last approximately one person-hour; flexibility is allowed to account for the heterogeneity in the size of houses. During the search, the inspector looks for live *T. infestans*, in addition to *T. infestans* eggs, exuviae, and feces. All inspectors participated in the study under a written informed consent approved under University of Pennsylvania IRB protocol number 824603 and Universidad Peruana Cayetano Heredia IRB protocol number 66427.

Inspectors using the paper maps did sometimes have access to two pieces of infestation risk information, as each house code (which is painted on the outside of the house) contains indicators

of participation in the attack phase of the vector control campaign, and a '+' sign at the end of the code if the house was ever found to be infested with *T. infestans*. This information is available only when standing directly in front of a house looking at the house code; it is not shown in the paper maps, meaning that inspectors using the paper maps could not see the spatial distribution of houses with these risk factors. In addition, many of the house codes have been painted over by homeowners in the years since the attack phase of the campaign, so all houses with these risk indicators are not identifiable by looking at the code.

Data analysis

To measure the effect of using the app in the field on inspector productivity we compared the total houses visited and total houses inspected between the app and paper maps. We also compared the proportion of total houses visited that ended up being inspected between the two mediums. We selected these metrics to test if the app was slowing the inspectors down or constricting their surveillance activities in some way, due to technical difficulties or otherwise. As mentioned above, we did not compare the number of infested houses found, since the prevalence of infestation is currently very low [29].

To investigate if inspectors used the risk information provided in the app to select houses to visit, we compared the proportion of houses visited that were higher risk houses (top two risk levels) when using the app and when using paper maps. To further investigate risk information use, we compared the proportion of total houses visits that resulted in inspection between the app and paper maps among just the higher risk houses visited (houses presented in the app as 'highest' and 'high risk') and just the lower risk houses visited (houses presented in the app as 'medium,' 'low,' and 'lowest' risk). We were interested in how the possession of information about a house's estimated risk level might influence the visit outcome (i.e., if a house was inspected, closed, inspection refused, an interview took place, or the inspector was asked to return at a later time).

Finally, we looked for qualitative evidence of risk information use by comparing maps of inspector movement patterns throughout the search zones when using the app and paper maps. The daily maps for each inspector are found in the VectorPoint repository (https://github.com/chirimacha/VectorPoint). We examined movement patterns on a smaller scale, such as changes in direction and the tendency to visit every neighboring house vs. skipping houses. In addition we looked at patterns on a larger scale, such as the tendency to visit all houses in one area of a zone versus visiting a few houses across several areas, as well as cumulative movement throughout the total search zone across all five days, which we refer to as 'spatial coverage' of the zone.

Statistical testing

To compare the total number of houses visited and houses inspected between the app and paper maps we used a paired t-test. For analyses involving house risk level, we split the houses visited into binary categories of higher and lower risk, consisting of the top two risk levels and the bottom three risk levels, respectively. For analyses involving visit outcome, we classified visit outcomes into the binary categories, 'inspection,' and 'other.' For all metrics except total houses visited and total house inspected, we carried out a preliminary analysis using Fisher's exact test to test for differences between the app and paper maps for each individual inspector, and a Mantel-Haenszel chi-square test with continuity correction to test for an overall shift in one direction among all inspectors. We carried out a second analysis using binomial Generalized Linear Mixed Models with "Inspector ID" random intercepts to test whether the inclusion of an "app" fixed effect helped to explain any variance. In this analysis, we compared the results from a model run with only inspector ID random intercepts with results from a model run with inspector ID random intercepts plus an app fixed effect. We used the BIC scores [87] as a metric to assess if the addition of the app

fixed effect improved the performance of the models, with smaller BIC scores suggesting model improvement. In addition, we evaluated the odds ratios in the app fixed effects models to understand if the app increased the odds of the outcome being evaluated. Models were fit by maximum likelihood (Laplace Approximation) using the 'glmr' function in the lme4 package [88] for R. All data analyses were carried out in the R statistical computing environment [82].

Results

Over the course of ten days, eight inspectors visited a total of 2,119 houses, of which 767 were inspected for *T. infestans* (Table 1). In the five days using the paper maps, 1,081 houses were visited, resulting in 366 inspections (33.9%). In five days with the app, 1,038 houses were visited, resulting in 401 inspections (38.6%).

Inspector	Total visits		Inspections	
	App	***Paper maps***	***App***	***Paper maps***
A	88	90	44	37
B	184	141	65	49
C	164	175	31	19
D	111	95	81	76
E	130	123	46	43
F	126	145	47	43
G	112	114	46	56
H	123	198	41	43
Total	1,038	1,081	401	366

Table 1. Columns: total number of house visits (left) and the number of house visits that resulted in inspection (right) using the app and paper maps. Rows: data for each inspector.

Effect of the app on inspector productivity

There was no difference between the app and paper maps in the total number of houses visited (paired t-test, p = 0.67, Table 1), the total number of houses inspected (paired t-test, p = 0.17, Table 1), or the proportion of total visits that resulted in inspection (Mantel-Haenszel test, chi square = 2.63, p = 0.105, odds ratio (OR) = 1.18, 95% CI = 0.97- 1.42; Figure 6), further suggesting that the app did not reduce productivity in the field. When looking at inspectors individually, no one inspected proportionally more houses with paper maps. Two inspectors, C and H, inspected a significantly higher proportion of houses when using the app as based on test p-values (Fisher's exact test, Inspector C: p = 0.046, OR= 1.91, 95% CI = 0.99-3.76; Inspector H: p = 0.026, OR = 1.80, 95% CI =1.05-3.08, Figure 6).

In the generalized linear mixed models (GLMMs), the odds ratio of the app fixed effect was 1.18 (95% CI = 0.97-1.42, Table 2), suggesting little effect of the app on the odds of a visited house being inspected. The standard deviation in inspector random effects was high regardless of whether or not the model included an app effect (SD = 0.76, 95% CI = 0.49-1.39 with the app; SD = 0.77, 95% CI = 0.49-1.40 without the app). The addition of the app fixed effect did not improve the model, with similar BIC scores for both models (138.71 and 138.80 for the model with and without the app, respectively). Results from the GLMMs are presented in Table 2.

Figure 6: Visit outcome distributions. Inspections shown in blue, visit outcomes making up the 'other' category shown in grey. *Significantly higher proportion of houses inspected when using the app (Fisher's exact test, p<0.05).

Risk information use

Five out of eight inspectors (B,D,E,F, and H; 62.5%, Figure 7), visited proportionally more higher risk houses (top two risk levels) when using the app than when using paper maps (Fisher's exact test, Inspector B: p = 4.14e-14, OR = 25.38, 95% CI = 7.80-131.06; Inspector D: p = 1.81e-14, OR=40.71, 95% CI = 9.94-360.00; Inspector E: p = 4.82e-07, OR = 4.42, 95% CI = 2.37-8.51; Inspector F: p < 2.2e-16, OR = 10.10, 95% CI = 5.58-18.80; Inspector H: p < 2.2e-16, OR = 14.86, 95% CI = 6.80-37.13). Two inspectors (25%), A and C, showed no difference in the proportion of total visits that were to higher risk houses when using the app (Fisher's exact test, Inspector A: p = 0.65, OR = 1.19, 95% CI = 0.63-2.25; Inspector C: p = 0.12, OR = 1.43, 95% CI = 0.90-2.26; Figure 7), and and one inspector, (G), visited proportionally more higher risk houses when using the paper map (Fisher's exact test, p < 2.2e-16, OR = 0.0, 95% CI = 0.0-0.02; Figure 7). Overall, there was a significant shift upward in the risk level of the houses visited from the paper maps to the app (Mantel-Haenszel test, chi-square = 104.44, p < 2.2e-16, common OR = 2.42, 95% CI = 2.00-2.92; Figure 7), suggesting that inspectors did incorporate the risk information provided in the app into their selection of houses to visit.

In the GLMM, the app effect was a significant predictor of a higher risk house being visited (p < 2e-16), and it increased the odds of visiting a higher risk house by 2.73 (95% CI = 2.24-3.32, Table 2). Standard deviation in inspector ID random effects was again high in both models (0.76, 95% CI = 0.48-1.37 with the app, and 0.77, 95% CI = 0.50-1.20 without the app). BIC scores suggested that the app improved the model, with BIC scores of 442.9 and 544.3 for the models with and without the app, respectively.

	Inspector productivity		Risk information use	
	Random inspector variation only	***With app fixed effect***	***Random inspector variation only***	***With app fixed effect***
Dependent variable	# inspections		# higher risk houses visited	
Predictor variable	Inspector variation	App use	Inspector variation	App use
Inspector variation intercept (95% CI)	-0.495 (-1.107-0.118)	-0.576 (-1.191-0.041)	0.571 (-0.031-1.177)	0.110 (-0.512-0.733)
Intercept SE	0.275	0.279	0.272	0.281
Intercept p-value	0.0723	0.0387	0.0356	0.697
App effect parameter estimate (95% CI)	n/a	0.163 (-0.026-0.352)	n/a	1.00 (0.808-1.202)
App effect parameter estimate SE	n/a	0.096	n/a	0.100
App effect p-value	n/a	0.0906	n/a	<2e-16
Estimated mean probability of outcome occurring (likelihood profile 95% CI)	0.379 (0.249-0.530)	0.398 (0.228- 0.597)	0.639 (0.492-0.765)	0.753 (0.573-0.874)
Random effects SD estimate (95% CI)	0.767 (0.489-1.395)	0.764 (0.487- 1.391)	0.756 (0.484-1.374)	0.774 (0.495-1.202)
Odds ratio (95% CI)	n/a	1.18 (0.974-1.142)	n/a	2.73 (2.240-3.321)
BIC score	138.80	138.71	544.30	442.90
Model improved with app fixed effect?	no		yes	

Table 2. Results for generalized linear mixed models investigating inspector productivity and risk information use with and without the Vectorpoint app as a fixed effect.

We did not observe any patterns associated with arm order (i.e, which medium was used in the first week and which in the second week, Figure 7). Out of the five inspectors who visited proportionally more higher risk houses using the app, two used the app first and three used the paper maps first. For the two inspectors who showed no difference between the app and paper maps, one inspector started with the app, and the other started with the paper map. The inspector who visited more higher risk houses with paper maps used the app first.

We also found no difference between the app and paper maps in the proportion of visits to higher risk houses or lower risk houses that resulted in inspection, suggesting that there is no association with knowledge of a house's infestation risk and obtaining permission to inspect it (Fisher's exact test, $p > 0.05$ for all inspectors for higher and lower risk houses; overall: Mantel-Haenszel test, higher risk houses: chi square = 1.71, $p = 0.19$, common OR = 1.19, 95% CI = 0.93-1.51; lower risk houses: chi square = 0.44, $p = 0.51$, common OR = 1.18, 95% CI = 0.78-1.78).

Figure 7. Infestation risk distribution of all houses visited for each inspector. Superscripts in the x axis text indicate arm order (i.e, which medium was used in the first week of the study, and which was used in the second week); values < 4% are not labeled due to space constraints; *p < 0.001, with more high risk houses visited using the app; ^p < 0.001, with more high risk houses visited using paper maps.

Qualitative characterization of movement patterns and spatial coverage

We observed two predominant movement patterns throughout the study. The first movement pattern was more individual house focused, while the second pattern was more focused on coverage of the total search zone. (As mentioned above, daily maps for each inspector are available in the VectorPoint respository, https://github.com/chirimacha/VectorPoint.) While most

inspectors consistently exhibited one pattern or the other, some inspectors displayed a mixture of both throughout the week. In the individual-house-focused movement pattern, inspectors tended to travel short distances (1-3 blocks/day), and visit every house on a block. Movement patterns appeared systematic and focused on covering all households in a small part of the zone, resulting in low spatial coverage of the total search zone. In the pattern that was more focused on search zone coverage, inspectors tended to travel longer distances, visiting a cluster of houses in one area, and then moving on to another part of the zone. Movement appeared less systematic than the individual-house-focused pattern, with a more holistic focus on the entire search zone, resulting in higher spatial coverage. There are many practical constraints in the field that will influence inspector movements, but larger movements across the zone were much more common in inspectors that tended toward the broader coverage of the search zone.

Seven out of eight inspectors (A, B,C,D,E,F and H) displayed at least one indication of risk information use in the context of these predominant movement patterns. In inspectors with individual house-focused movement patterns, risk information use was indicated when they skipped lower risk houses in between higher risk houses, or made abrupt changes in direction to avoid a cluster of lower risk houses, which sometimes resulted in increased spatial coverage of the search zone. While using the paper maps, these inspectors rarely skipped houses or changed direction, and their spatial coverage of the zone was low. Inspectors who were more focused on coverage of the total zone tended to indicate risk information use with movement that was more systematic and directed toward higher risk houses when using the app. When using the app in areas with clusters of higher risk houses, some of these inspectors adjusted their usual broader coverage movement pattern to be more individual house focused, resulting in visits to every individual house on a higher risk block.

We observed indications of risk information use in inspectors that did visit significantly more houses with the app (B,D,E,F,H) and in those that did not (A and C). Inspector C did not visit significantly more high risk houses, but did skip clusters of lower risk houses on some days, suggesting they were 'trying out' the risk information. Inspector A greatly increased spatial coverage when using the app as compared to the paper map, which was due to a movement pattern that appeared to 'sample' houses from a new risk level each day. Over the week, the inspector progressed in order of risk level from visiting lowest risk houses on day 1 to visiting highest risk on day 5. This inspector did not visit significantly more houses using the app, but did appear to be paying attention to the risk information in the app. Indeed, by the end of the week, the inspector was skipping houses that were low and lowest risk.

Only one inspector (G) showed no qualitative or quantitative indications of using the risk information in the app to select houses to visit. In fact, this inspector unexpectedly visited significantly more higher risk houses when using the paper map, which we attribute to clustering of houses of the same risk level in the search zone. Based on examination of this inspector's movement patterns when using the paper map, it seems that this occurred because the inspector displayed a strong tendency toward fine-scale (individual house-focused) movement, and they selected their day one starting point based on house code, (lowest codes first), and proceeding through the zone in numerical order. Coincidentally, the starting house was located in the beginning of a large cluster of higher risk houses, resulting in only higher risk houses being visited with the paper map.

Discussion

Here we present a mobile app designed to provide the opportunity to incorporate data-based risk information into field surveillance for the Chagas disease vector species *Triatoma infestans* in the

572 city of Arequipa, Perú. In the study comparing surveillance using the app to the current practice of
573 surveillance using paper maps, we observed multiple indications that the risk information provided
574 in the app was used in the selection of houses to inspect, suggesting that the app is a feasible tool to
575 enhance vector surveillance and support decision making in the field.
576
577 *mHealth state of the art*
578 mHealth tools, defined as mobile and wireless technologies for health-related objectives [89] are
579 being introduced at a rapid-fire pace [90]. While the majority of mHealth apps are for personal use
580 relating to aspects of individual health, the use of apps for disease surveillance in resource-limited
581 settings has been growing steadily. The large number of data collection apps available for disease
582 surveillance has provided the opportunity to replace often slow and cumbersome paper-based data
583 collection systems with mobile data collection systems in which data are sent directly to a central
584 database at the time of collection. Ideally, the shift from paper to technology will lead to increased
585 data completeness and coverage in resource-limited settings, a critical step for achieving disease
586 surveillance goals [91]. (Of course, new technologies bring new challenges, some of which are
587 detailed below.) The next step is now integrating data collection functionalities with more complex
588 tools to support ongoing decision making in the field in real or close to real time. While several data
589 collection apps offer basic data visualization capabilities such as simple maps or bar graphs
590 showing raw data distributions, these functionalities are often not immediately available to the
591 individuals collecting data in the field, and the usefulness of raw data for on the ground decision
592 making can be limited. The VectorPoint app is among the first in its marriage of data collection with
593 predictive modeling and spatial data visualization, all of which are intended to support the
594 individual collecting the data
595
596 *Risk information use*

As mentioned previously, for VectorPoint to meet the objective of integrating data into vector surveillance activities, inspectors with years of experience and well-established routines must adapt their decision-making processes to include at least some of the risk information in the app. Model-generated estimates are not necessarily more valuable than knowledge derived from experience, and VectorPoint is meant to complement an inspector's experience, not replace it. In the study, we observed multiple indications that risk information was used by the inspectors. Namely, we observed a significant upward shift among all inspectors in the number of high risk houses visited when using the app compared to paper maps, and a majority of the inspectors visited significantly more higher risk houses when using the app. These results were then confirmed in a second analysis comparing generalized linear mixed models run with and without the app, in which the the app was a highly significant predictor of a higher risk house being visited. In the maps of inspector movement throughout the search zones, we also observed changes in predominant movement patterns when using the app that indicated risk information use. Even inspectors who did not visit significantly more higher risk houses with the app appeared to at least 'try out' targeting higher risk houses or explore the different risk levels, suggesting that over time inspectors may gradually introduce the risk information into their surveillance activities.

While our results are encouraging, one inspector did not show any signs of using the risk information, and we cannot assume adoption would be universal. Indeed, our more rigorous analyses using generalized linear mixed models confirmed that random among-inspector variation is high regardless of the outcome tested. Willingness to use the information in the app may be associated with factors specific to the individual, such as age or experience with technology [89], and we may need to take into account inspector demographic information when analyzing app use in future studies in order to develop strategies for increasing adoption. In addition, motivation to use the app may increase as more data are collected, allowing us to compare infestation indices

between houses inspected with the app versus paper maps. As mentioned above, this study was not scaled to formally compare infestation indices; only one infested house was found in the study, and the inspector was using a paper map at the time. Specialized training or more creative solutions such as incentives [92–94] or games [95,96] may be necessary in cases where inspectors have trouble engaging with the app, or if enthusiasm for the new technology wanes over time.

Spatial coverage of surveillance zones

Although the tendency to travel or not tended to be a fixed characteristic between both arms of the study, in a few cases we observed qualitative changes in spatial coverage of the search zone when using the app. While the app is not currently designed to directly address the issue of spatial coverage, the model is expected to redistribute the relative risk estimates throughout the search zone every time it is run with new inspection data. In other words, if a house is surveyed and found to be not infested with *T. infestans*, infestation risk should decrease for all households within a certain distance of the uninfested house. In this way, inspectors who use the risk information in the app to target high risk houses may also display even spatial coverage of their search zone. In the study, we observed this effect only occasionally, and in some cases, inspectors targeting high risk houses displayed reduced spatial coverage of their search zone when higher risk houses were clustered. This outcome is likely due to the relatively small amount of data collected by each inspector, and the varying amounts of information available in each search area prior to the study. As detailed above, we expect risk level clustering to be dynamic, and, in the absence of a positive house, risk distribution should even out as more data are collected. The utility of the model behind the app, like all Bayesian models, increases as more data is accumulated.

Extensions to the app, or the model itself, may be useful in improving spatial coverage of surveillance, and for exploring more complex questions revolving around the benefits or drawbacks

of using the app. In particular, we do not know if a trade off occurs between exploration and exploitation when using the app; by focusing on areas that are known or predicted to have a higher risk of infestation (i.e., exploitation), will we fail to detect new high risk foci (i.e., exploration)?

Feasibility: overcoming logistical barriers and increasing adoption

The use of mHealth tools in resource-limited settings presents inherent challenges [50,55,73] due to the financial and technical requirements of cell-phone based systems. An advantage of VectorPoint is that it uses entirely free open-source technologies, which allows large-scale deployment at little cost. The remaining operational costs involve the acquisition of mobile devices and their corresponding cellular network subscriptions. For cloud-based and web-based apps such as VectorPoint, these networks must provide data coverage of at least 2G speeds. In our field tests of VectorPoint, we found that frequent interruption and reloading of the maps were avoided when data speeds were 3G or higher. Fortunately, Peru has a robust mobile phone culture, with 117 mobile phone subscriptions per 100 people in 2016 [97], higher than the world average of 101.5. There are several networks available throughout Peru, all offering high-speed data plans. In cases where network problems do occur, VectorPoint has a caching functionality, as mentioned above, that allows data entry, although some internet connectivity is still needed for efficient use of the app. We are working towards a completely offline-capable version of VectorPoint to overcome this limitation. In our study, there was no difference in the number of houses visited between paper maps and the app, suggesting that any connectivity issues encountered did not significantly slow surveillance activities.

Being a cloud-based computing app means that VectorPoint can be run on any device without any configuration or installation, but it does require human expertise to oversee and problem-solve its backend components. Fortunately, there is a growing culture and acceptance of electronic health

672 tools in Peru, which has been found to contribute to the sustainability of health information
673 systems in resource-limited settings [73]. Gozzer Infante [98] reviewed 38 electronic health
674 systems that were introduced in Peru between 2002-2010, and 66% of them were still being used
675 in 2015. These systems include Alerta, an large multi-disease top down surveillance system [99]
676 Netlab, a laboratory support system for HIV treatment [100] and Magpi, a data collection app being
677 used by researchers to collect data on HPV prevalence [61]. As of 2018, several mHealth apps were
678 confirmed to be in use at the national level, (Leonardo Rojas, Peruvian Institute of Health, personal
679 communication), such as Guardianes de la Salud [101], a mobile phone based disease surveillance
680 system for use during the early 2018 Papal visit. In the case of VectorPoint, we collaborated with
681 local personnel in the development of the app from the start, and they currently control both its
682 operational use and its engineering. However, it should be mentioned that although the VectorPoint
683 app is flexible, the predictive model must still be adapted to fit the local epidemiological and
684 ecological characteristics of each disease surveillance scenario to which it is applied, which could
685 present challenges in cases where expertise is limited.

686 **Conclusion**

687 Our findings suggest that the VectorPoint app could be a useful tool to integrate evidence and
688 models into epidemiological surveillance in cities. The app was designed to be used for *T. infestans*
689 surveillance, but its components are cloud-based, open-source, and ready to be adapted to other
690 appropriate scenarios, although the availability of sufficient data and/or resources may be a hurdle
691 to overcome in some cases. VectorPoint is simple to use, but critical in function: without staying
692 vigilant to remaining and re-emerging vector foci following a vector control campaign, disease
693 transmission inevitably returns and progress achieved is reversed [102].

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References

1. Who. Chagas Disease Fact Sheet [Internet]. Geneva, Switzerland: World Health Organization; 2017. Available: http://www.who.int/mediacentre/factsheets/fs340/en/
2. Vos T, Barber RM, Bell B, Bertozzi-Villa A, Biryukov S, Bolliger I, et al. Global, regional, and national incidence, prevalence, and years lived with disability for 301 acute and chronic diseases and injuries in 188 countries, 1990-2013: A systematic analysis for the Global Burden of Disease Study 2013. Lancet. 2015;386: 743–800.
3. World Health Organization. Chagas disease in Latin America: an epidemiological update based on the 2010 estimates. Weekly epidemiological record. 2015;6: 33–44.
4. Bern C. Chagas' Disease. N Engl J Med. 2015;373: 456–466.
5. Rassi A Jr, Rassi A, Marin-Neto JA. Chagas disease. Lancet. Elsevier Ltd; 2010;375: 1388–1402.
6. Dias E. Estudos sobre o *Schizotrypanum cruzi*. Mem Inst Oswaldo Cruz. 1934;28: 1–110.
7. Gorla D, Ponce C, Dujardin JP, Schofield CJ. Control strategies against Triatominae. In: Telleria J, Tibayrenc M, editors. American Trypanosomiasis Chagas Disease. 1st ed. Elsevier; 2010. pp. 233–245.
8. Hashimoto K, Schofield CJ. Elimination of *Rhodnius prolixus* in Central America. Parasit Vectors. BioMed Central Ltd; 2012;5: 45.
9. Schofield CJ, Dias JCP. The Southern Cone Initiative against Chagas Disease. Adv Parasitol. 1999;42: 1–27.
10. Coura JR, Viñas PA. Chagas disease: a new worldwide challenge. Nature. 2010;465: S6–S7.
11. Zeledon R. Chagas Disease: An Ecological Appraisal with Special Emphasis on its Insect Vectors. Annu Rev Entomol. 1981;26: 101–133.

12. Corredor Arjona A, Alvarez Moreno CA, Agudelo CA, Bueno M, López MC, Cáceres E, et al. Prevalence of *Trypanosoma cruzi* and *Leishmania chagas*i infection and risk factors in a Colombian indigenous population. Rev Inst Med Trop Sao Paulo. 1999;41: 229–234.

13. Feliciangeli MD, Sánchez-Martín MJ, Suárez B, Marrero R, Torrellas A, Bravo A, et al. Risk factors for *Trypanosoma cruzi* human infection in Barinas State, Venezuela. Am J Trop Med Hyg. 2007;76: 915–921.

14. Lardeux F, Depickère S, Aliaga C, Chavez T, Zambrana L. Experimental control of *Triatoma infestans* in poor rural villages of Bolivia through community participation. Trans R Soc Trop Med Hyg. 2014;109: 150–158.

15. Medina-Torres I, Vázquez-Chagoyán JC, Rodríguez-Vivas RI, de Oca-Jiménez RM. Risk factors associated with triatomines and its infection with *Trypanosoma cruzi* in rural communities from the southern region of the State of Mexico, Mexico. Am J Trop Med Hyg. 2010;82: 49–54.

16. Bustamante DM, De Urioste-Stone SM, Juarez JG, Pennington PM. Ecological, social and biological risk factors for continued *Trypanosoma cruzi* transmission by *Triatoma dimidiata* in Guatemala. PLoS One. 2014;9. doi:10.1371/journal.pone.0104599

17. Gurtler RE, Cohen JE, Cecere MC, Lauricella MA, Chuit R, Segura EL. Influence of humans and domestic animals on the household prevalence of *Trypanosoma cruzi* in *Triatoma infestans* populations in northwest Argentina. Am J Trop Med Hyg. 1998;58: 748–758.

18. Coura JR. Chagas disease: what is known and what is needed A background article. Mem Inst Oswaldo Cruz. 2007;102: 113–122.

19. Ricardo-Silva AH, Lopes CM, Ramos LB, Marques WA, Mello CB, Duarte R, et al. Correlation between populations of Rhodnius and presence of palm trees as risk factors for the emergence of Chagas disease in Amazon region, Brazil. Acta Trop. 2012;123: 217–223.

20. Carrasco HJ, Segovia M, Londoño JC, Ortegoza J, Rodríguez M, Martínez CE. *Panstrongylus geniculatus* and four other species of triatomine bug involved in the *Trypanosoma cruzi* enzootic cycle: high risk factors for Chagas' disease transmission in the Metropolitan District of Caracas, Venezuela. Parasit Vectors. 2014;7: 1003–1028.

21. Gajate P, Pietrokovsky S, Abramo Orrego L, Pérez O, Monte A, Belmonte J, et al. *Triatoma infestans* in Greater Buenos Aires, Argentina. Mem Inst Oswaldo Cruz. 2001;96: 473–477.

22. Guzman-Tapia Y, Ramírez-Sierra MJ, Dumonteil E. Urban Infestation by *Triatoma dimidiata* in the City of Mérida, Yucatán, México. Vector-Borne and Zoonotic Diseases. 2007;7: 597–606.

23. Levy MZ, Bowman NM, Kawai V, Waller LA, Geny J, Benzaquen EC, et al. Periurban *Trypanosoma cruzi*-infected *Triatoma infestans*, Arequipa , Peru. Emerg Infect Dis. 2006;12: 1345–1352.

24. Medrano-Mercado N, Ugarte-Fernandez R, Butrón V, Uber-Busek S, Guerra HL, De Araújo-Jorge TC, et al. Urban transmission of Chagas disease in Cochabamba, Bolivia. Mem Inst Oswaldo Cruz. 2008;103: 423–430.

25. Ricardo-Silva A, Monte-Gonçalves TC, Luitgards-Moura JF, Macedo-Lopes C, Pedrosa-da-Silva S, Queiroz-Bastos A, et al. *Triatoma maculata* colonises urban domicilies in Boa Vista, Roraima,

763 Brazil. Mem Inst Oswaldo Cruz. 2016;111: 703–706.

764 26. Rodrigues VLCC, Pauliquevis Junior C, da Silva RA, Wanderley DMV, Guirardo MM, Rodas LAC,
765 et al. Colonização de palmeiras por *Rhodnius neglectus* e invasão domiciliar em área urbana,
766 Araçatuba, São Paulo, Brasil. Rev Inst Med Trop Sao Paulo. 2014;56: 213–218.

767 27. Vallvé SL, Rojo H, Wisnivesky-Colli C. Urban ecology of *Triatoma infestans* in San Juan,
768 Argentina. Mem Inst Oswaldo Cruz. 1996;91: 405–408.

769 28. Levy MZ, Barbu CM, Castillo-Neyra R, Quispe-Machaca VR, Ancca-Juarez J, Escalante-Mejia P, et
770 al. Urbanization, land tenure security and vector-borne Chagas disease. Proceedings of the
771 Royal Society B: Biological Sciences. 2014;281: 20141003–20141003.

772 29. Barbu CM, Hong A, Manne JM, Small DS, Quintanilla Calderón JE, Sethuraman K, et al. The
773 Effects of City Streets on an Urban Disease Vector. PLoS Comput Biol. 2013;9.
774 doi:10.1371/journal.pcbi.1002801

775 30. Barbu CM, Buttenheim AM, Hancco Pumahuanca ML, Quintanilla Calderón JE, Salazar R,
776 Carrión M, et al. Residual infestation and recolonization during urban Triatoma infestans bug
777 control campaign, Peru. Emerg Infect Dis. 2014;20: 2055–2063.

778 31. Bayer AM, Hunter GC, Gilman RH, Cornejo Del Carpio JG, Naquira C, Bern C, et al. Chagas
779 disease, migration and community settlement patterns in Arequipa, Peru. PLoS Negl Trop Dis.
780 2009;3: e567.

781 32. Bowman NM, Kawai V, Levy MZ, Cornejo del Carpio JG, Cabrera L, Delgado F, et al. Chagas
782 disease transmission in periurban communities of Arequipa, Peru. Clin Infect Dis. 2008;46:
783 1822–1828.

784 33. Buttenheim AM, Paz-Soldan V, Barbu C, Skovira C, Quintanilla Calderón J, Mollesaca Riveros
785 LM, et al. Is participation contagious? Evidence from a household vector control campaign in
786 urban Peru. J Epidemiol Community Health. 2014;68: 103–109.

787 34. Delgado S, Ernst KC, Pumahuanca ML, Yool SR, Comrie AC, Sterling CR, et al. A country bug in
788 the city: urban infestation by the Chagas disease vector *Triatoma infestans* in Arequipa, Peru.
789 Int J Health Geogr. 2013;12: 48.

790 35. Levy MZ, Kawai V, Bowman NM, Waller L a., Cabrera L, Pinedo-Cancino VV, et al. Targeted
791 screening strategies to detect *Trypanosoma cruzi* infection in children. PLoS Negl Trop Dis.
792 2007;1: e103.

793 36. Levy MZ, Small DS, Vilhena DA, Bowman NM, Kawai V, Cornejo del Carpio JG, et al. Retracing
794 Micro-Epidemics of Chagas Disease Using Epicenter Regression. PLoS Comput Biol. 2011;7.
795 doi:10.1371/journal.pcbi.1002146

796 37. Paz-Soldán VA, Bauer KM, Hunter GC, Castillo-Neyra R, Arriola VD, Rivera-Lanas D, et al. To
797 spray or not to spray? Understanding participation in an indoor residual spray campaign in
798 Arequipa, Peru. Glob Public Health. 2016;1692: 1–18.

799 38. Banks K, McDonald SM, Scialom F. Mobile Technology and the Last Mile: "Reluctant Innovation"
800 and FrontlineSMS. Innovations: Technology, Governance, Globalization. MIT Press; 2011;6: 7–
801 12.

802 39. Banks K. FrontlineSMS-A Text Messaging System for NGOs. Oryx. 2007;40: 17.

803 40. Banks K, Hersman E. FrontlineSMS and Ushahidi - a demo. 2009 International Conference on
804 Information and Communication Technologies and Development (ICTD). 2009. pp. 484–484.

805 41. Malawi U. RapidSMS Technology. Retrieved June. 2009;

806 42. Blaschke S. An assessment of the use of RapidSMS to monitor the Hepatitis-E outbreak in
807 northern uganda.

808 43. UNICEF's U-Report: using mobile technology for youth participation in policymaking. In:
809 Center for Public Impact [Internet]. 22 Mar 2016 [cited 22 Feb 2018]. Available:
810 https://www.centreforpublicimpact.org/case-study/unicef-ureport/

811 44. U-Report [Internet]. [cited 22 Feb 2018]. Available: http://ureport.ug/

812 45. Okolloh O. Ushahidi, or "testimony": Web 2.0 tools for crowdsourcing crisis information.
813 Participatory learning and action. IIED-Participatory Learning and Action; 2009;59: 65–70.

814 46. Puleio M, Jennings V. CycleTel in India: From Proof-of-Concept to Deployment. Presentation to
815 mHealth Working Group. 2011; Available:
816 http://coregroup.secure.nonprofitsoapbox.com/storage/CycleTel_mHealth_WG_Jan2011-1.pdf

817 47. Rai RK. Tracking women and children in a Continuum of Reproductive, Maternal, Newborn,
818 and Child Healthcare (RMNCH) in India. J Epidemiol Glob Health. 2014;4: 239–243.

819 48. Lian L. GeoChat on iPhone : A Map-based Chat Application on iPhone [Internet]. 2010.
820 Available: http://www.diva-portal.org/smash/record.jsf?pid=diva2:351488

821 49. Yang C, Yang J, Luo X, Gong P. Use of mobile phones in an emergency reporting system for
822 infectious disease surveillance after the Sichuan earthquake in China. Bull World Health Organ.
823 2009;87: 619–623.

824 50. Déglise C, Suzanne Suggs L, Odermatt P. SMS for disease control in developing countries: A
825 systematic review of mobile health applications. J Telemed Telecare. 2012;18: 273–281.

826 51. Lewis SL, Feighner BH, Loschen WA, Wojcik RA, Skora JF, Coberly JS, et al. Sages: A suite of
827 freely-available software tools for electronic disease surveillance in resource-limited settings.
828 PLoS One. 2011;6: 1–7.

829 52. Hartung C, Lerer A, Anokwa Y, Tseng C, Brunette W, Borriello G. Open Data Kit: Tools to Build
830 Information Services for Developing Regions. Proceedings of the 4th ACM/IEEE International
831 Conference on Information and Communication Technologies and Development. New York, NY,
832 USA: ACM; 2010. pp. 18:1–18:12.

833 53. Tom-Aba D, Olaleye A, Olayinka AT, Nguku P, Waziri N, Adewuyi P, et al. Innovative
834 Technological Approach to Ebola Virus Disease Outbreak Response in Nigeria Using the Open
835 Data Kit and Form Hub Technology. PLoS One. 2015;10: e0131000.

836 54. Mwabukusi M, Karimuribo ED, Rweyemamu MM, Beda E. Mobile technologies for disease
837 surveillance in humans and animals. Onderstepoort J Vet Res. 2014;81: 1–5.

838 55. Aanensen DM, Huntley DM, Feil EJ, Al-Own F 'a, Spratt BG. EpiCollect: Linking smartphones to
839 web applications for epidemiology, ecology and community data collection. PLoS One. 2009;4.
840 doi:10.1371/journal.pone.0006968

841 56. Karimuribo ED, Sayalel K, Beda E, Short N, Wambura P, Mboera LG, et al. Towards One Health
842 disease surveillance: The Southern African Centre for Infectious Disease Surveillance
843 approach. Onderstepoort J Vet Res. 2012;79: 1–7.

844 57. Tumwebaze H, Tumwesigye E, Baeten JM, Kurth AE, Revall J, Murnane PM, et al. Household-
845 Based HIV Counseling and Testing as a Platform for Referral to HIV Care and Medical Male
846 Circumcision in Uganda: A Pilot Evaluation. PLoS One. 2012;7.
847 doi:10.1371/journal.pone.0051620

848 58. Bollinger RC, McKenzie-White J, Gupta A. Building a global health education network for
849 clinical care and research. The benefits and challenges of distance learning tools. Lessons
850 learned from the Hopkins Center for Clinical Global Health Education. Infect Dis Clin North Am.
851 2011;25: 385–398.

852 59. Medic Mobile [Internet]. [cited 22 Feb 2018]. Available: https://medicmobile.org/

853 60. Magpi [Internet]. [cited 22 Feb 2018]. Available: https://home.magpi.com/

854 61. Blas MM, Brown B, Menacho L, Alva IE, Silva-Santisteban A, Carcamo C. HPV Prevalence in
855 Multiple Anatomical Sites among Men Who Have Sex with Men in Peru. PLoS One. 2015;10:
856 e0139524.

857 62. Mahmud MA, Spigt M, Mulugeta Bezabih A, López Pavon I, Dinant G-J, Blanco Velasco R. Risk
858 factors for intestinal parasitosis, anaemia, and malnutrition among school children in Ethiopia.
859 Pathog Glob Health. 2013;107: 58–65.

860 63. Human Network International. DataWinners: Data Collection for Development [Internet].
861 [cited 22 Feb 2018]. Available: https://www.datawinners.com/

862 64. WebFirst. PhiCollect [Internet]. [cited 22 Feb 2018]. Available:
863 https://www.webfirst.com/phicollect

864 65. Mtema Z, Changalucha J, Cleaveland S, Elias M, Ferguson HM, Halliday JEB, et al. Mobile Phones
865 As Surveillance Tools: Implementing and Evaluating a Large-Scale Intersectoral Surveillance
866 System for Rabies in Tanzania. PLoS Med. 2016;13: 1–12.

867 66. Haque F, Ball RL, Khatun S, Ahmed M, Kache S, Chisti MJ, et al. Evaluation of a Smartphone
868 Decision-Support Tool for Diarrheal Disease Management in a Resource-Limited Setting. PLoS
869 Negl Trop Dis. 2017;11: 1–19.

870 67. Robertson C, Sawford K, Daniel SLA, Nelson TA, Stephen C. Mobile phone-based infectious
871 disease surveillance system, Sri Lanka. Emerg Infect Dis. 2010;16: 1524–1531.

872 68. Garcia Sylim P, Cristina S-AC. Development of a low-cost electronic data collection tool for a
873 health facility survey study: lessons learned in the field. Journal of the international society for
874 telemedicine and ehealth. 2016;4: 1–8.

875 69. GIS Cloud Mobile Data Collection [Internet]. [cited 22 Feb 2018]. Available:

876 https://www.giscloud.com/apps/mobile-data-collection

877 70. Coloma J, Suazo H, Harris E, Holston J. Dengue chat: A novel web and cellphone application
878 promotes community-based mosquito vector control. Annals of global health. 2016;82: 451.

879 71. Lozano-Fuentes S, Wedyan F, Hernandez-Garcia E, Sadhu D, Ghosh S, Bieman JM, et al. Cell
880 phone-based system (Chaak) for surveillance of immatures of dengue virus mosquito vectors. J
881 Med Entomol. 2013;50: 879–889.

882 72. Bolivia: SMS contra el Chagas. In: Medicos Sin Fronteras [Internet]. [cited 22 Feb 2018].
883 Available: https://www.msf.es/actualidad/bolivia/bolivia-sms-chagas

884 73. Kimaro HC, Nhampossa JL. Analyzing the problem of unsustainable health information systems
885 in less-developed economies: Case studies from Tanzania and Mozambique. Information
886 Technology for Development. 2005;11: 273–298.

887 74. Brewer CA, Hatchard GW, Harrower MA. ColorBrewer in print: A catalog of color schemes for
888 maps. Cartogr Geogr Inf Sci. 2003;30: 5–32.

889 75. Brewer CA, MacEachren AM, Pickle LW, Herrmann D. Mapping mortality: Evaluating color
890 schemes for choropleth maps. Ann Assoc Am Geogr. 1997;87: 411–438.

891 76. Brewer CA, Pickle L. Evalutation of methods for classifying Epidemiological data on choropleth
892 maps in series. Ann Assoc Am Geogr. 2002;92: 662–681.

893 77. Smith-Jackson TL, Wogalter MS. Users' Hazard Perceptions of Warning Components: An
894 Examination of Colors and Symbols. Proc Hum Fact Ergon Soc Annu Meet. SAGE Publications
895 Inc; 2000;44: 6–55–6–58.

896 78. Rasmussen CE, Williams CKI. Gaussian Processes for Machine Learning. MIT Press; 2006.

897 79. Lindgren F, Rue H, Lindström J. An explicit link between Gaussian fields and Gaussian Markov
898 random fields: the stochastic partial differential equation approach. J R Stat Soc Series B Stat
899 Methodol. 2011;73: 423–498.

900 80. Song H-R, Fuentes M, Ghosh S. A comparative study of Gaussian geostatistical models and
901 Gaussian Markov random field models1. J Multivar Anal. 2008;99: 1681–1697.

902 81. Lindgren F, Rue H. Bayesian Spatial Modelling with R - INLA. J Stat Softw. 2015;63.
903 doi:10.18637/jss.v063.i19

904 82. R.Core.Team. R: A language and environment for statistical computing. Vienna, Austria: R
905 Foundation for Statistical Computing; 2017.

906 83. Rose EB, Lee K, Roy JA, Small D, Ross ME, Castillo-Neyra R, et al. Risk maps for cities:
907 Incorporating streets into geostatistical models. Spat Spatiotemporal Epidemiol. 2018;
908 doi:10.1016/j.sste.2018.08.003

909 84. Ooms J, James D, DebRoy S, Wickham H, Horner J. RMySQL: Database Interface and "MySQL"
910 Driver for R. 2017.

911 85. Chang W, Cheng J, Allaire JJ, Xie Y, McPherson J. shiny: Web Application Framework for R

912 [Internet]. 2017. Available: https://cran.r-project.org/package=shiny

913 86. Cheng J, Karambelkar B, Xie Y. leaflet: Create Interactive Web Maps with the JavaScript
914 "Leaflet" Library [Internet]. 2017. Available: https://cran.r-project.org/package=leaflet

915 87. Schwarz G. Estimating the Dimension of a Model. Ann Stat. Institute of Mathematical Statistics;
916 1978;6: 461–464.

917 88. Bates D, Mächler M, Bolker B, Walker S. Fitting Linear Mixed-Effects Models Using lme4.
918 Journal of Statistical Software, Articles. 2015;67: 1–48.

919 89. Geneva: World Health Organization. The MAPS toolkit: mHealth Assessment and Planning for
920 Scale. 2015.

921 90. Becker S, Miron-Shatz T, Schumacher N, Krocza J, Diamantidis C, Albrecht U-V. mHealth 2.0:
922 Experiences, Possibilities, and Perspectives. JMIR mHealth and uHealth. 2014;2: e24.

923 91. Lescano AG, Larasati RP, Sedyaningsih ER, Bounlu K, Araujo-Castillo RV, Munayco-Escate CV, et
924 al. Statistical analyses in disease surveillance systems. BMC Proc. 2008;2: S7.

925 92. Behrman JR, Gallardo-García J, Parker SW, Todd PE, Vélez-Grajales V. Are Conditional Cash
926 Transfers Effective in Urban Areas? Evidence from Mexico. Educ Econ. 2012;20: 233–259.

927 93. Van Herck P, De Smedt D, Annemans L, Remmen R, Rosenthal MB, Sermeus W. Systematic
928 review: Effects, design choices, and context of pay-for-performance in health care. BMC Health
929 Serv Res. 2010;10: 247.

930 94. Emanuel EJ, Ubel PA, Kessler JB, Meyer G, Muller RW, Navathe AS, et al. Using Behavioral
931 Economics to Design Physician Incentives That Deliver High-Value Care. Ann Intern Med.
932 2016;164: 114–119.

933 95. Garard DL, Lippert L, Hunt SK, Paynton ST. Alternatives to traditional instruction: Using games
934 and simulations to increase student learning and motivation. Commun Res Rep. Routledge;
935 1998;15: 36–44.

936 96. Clark AM, Clark MTG. Pokémon Go and Research: Qualitative, Mixed Methods Research, and the
937 Supercomplexity of Interventions. International Journal of Qualitative Methods. SAGE
938 Publications Inc; 2016;15: 1609406916667765.

939 97. The World Bank Group. Mobile cellular subscriptions (per 100 people) | Data [Internet]. [cited
940 26 Feb 2018]. Available:
941 https://data.worldbank.org/indicator/IT.CEL.SETS.P2?end=2016&start=2016&view=map

942 98. Gozzer Infante E. Una visión panorámica de las experiencias de telesalud en Perú. Rev Peru
943 Med Exp Salud Publica. 2015;32: 385–390.

944 99. Soto G, Araujo-Castillo RV, Neyra J, Fernandez M, Leturia C, Mundaca CC, et al. Challenges in the
945 implementation of an electronic surveillance system in a resource-limited setting: Alerta, in
946 Peru. BMC Proc. 2008;2 Suppl 3: S4.

947 100. García PJ, Vargas JH, Caballero PN, Calle JV, Bayer AM. An e-health driven laboratory
948 information system to support HIV treatment in Peru: E-quity for laboratory personnel, health

providers and people living with HIV. BMC Med Inform Decis Mak. 2009;9: 1–11.

101. Leal-Neto OB, Dimech GS, Libel M, Oliveira W, Ferreira JP. Digital disease detection and participatory surveillance: overview and perspectives for Brazil. Rev Saude Publica. 2016;50: 17.

102. Delgado S, Neyra RC, Machaca VRQ, Juárez JA, Chu LC, Verastegui MR, et al. A history of Chagas disease transmission, control, and re-emergence in peri-rural La Joya, Peru. PLoS Negl Trop Dis. 2011;5. doi:10.1371/journal.pntd.0000970

Supporting information Legends

S1. Study data. Data collected in the study comparing surveillance using the VectorPoint app and paper maps.

Figure 1

Click here to access/download;Figure;Fig1.tif

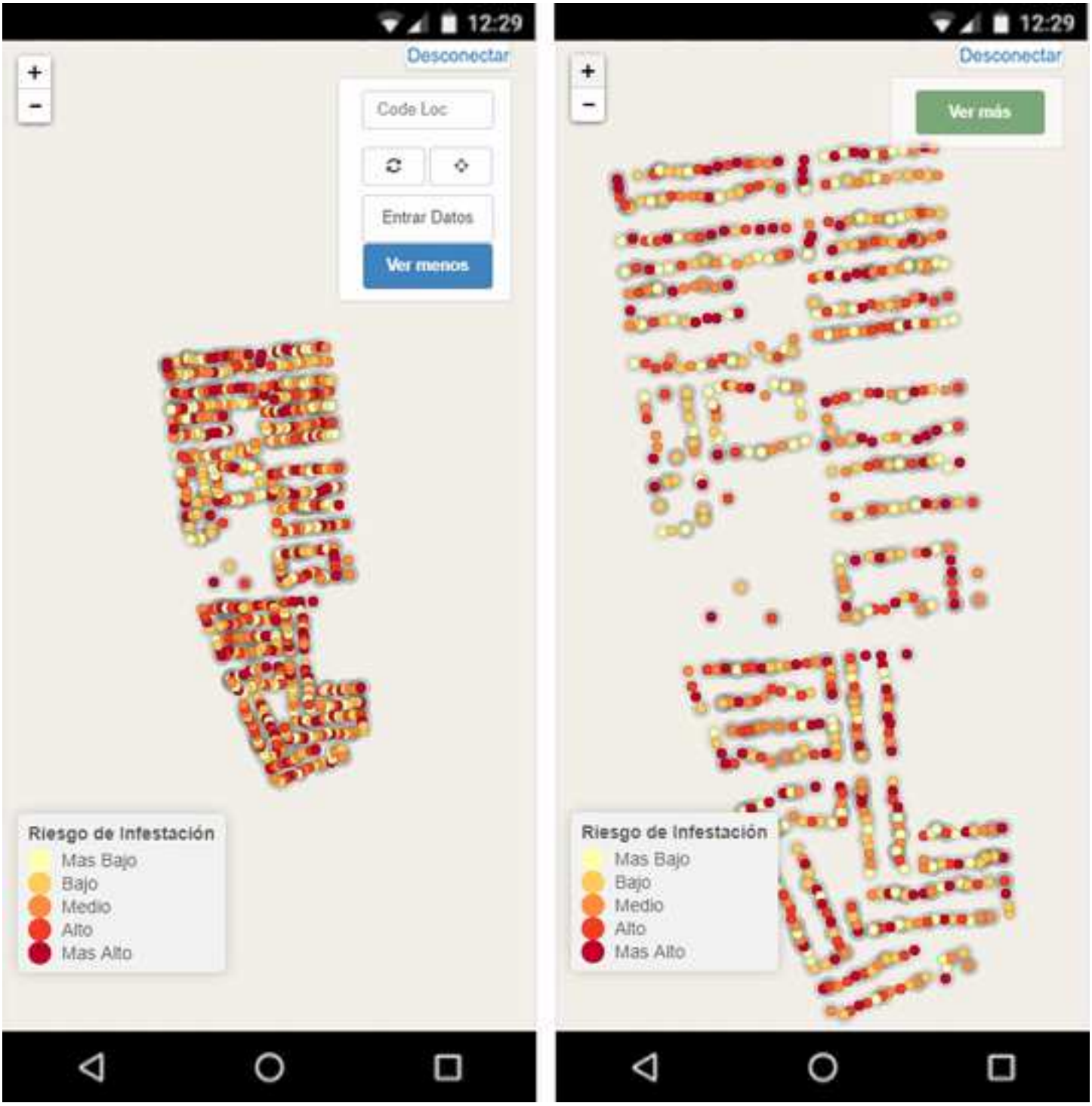


Figure 2

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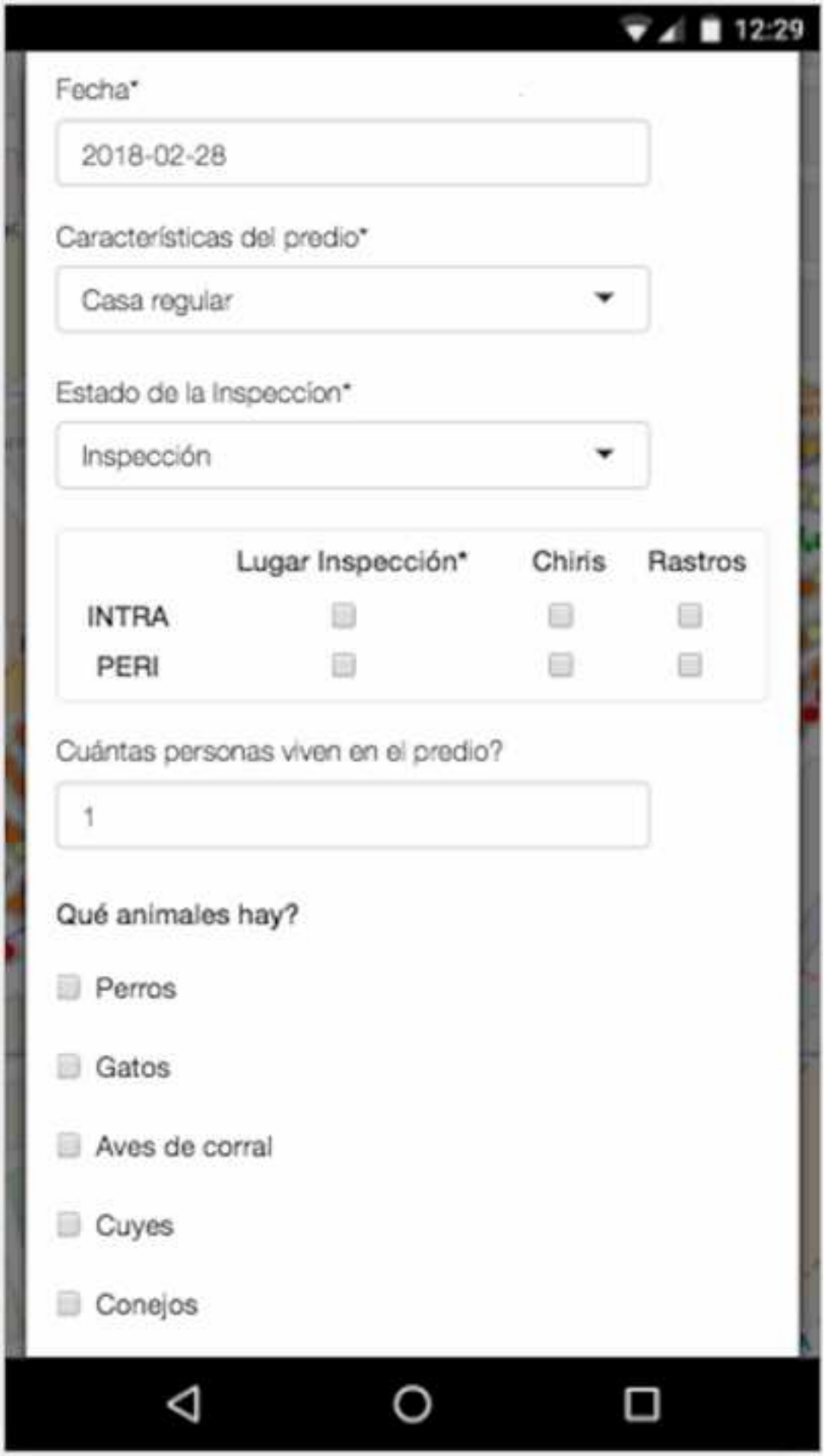


Figure 3

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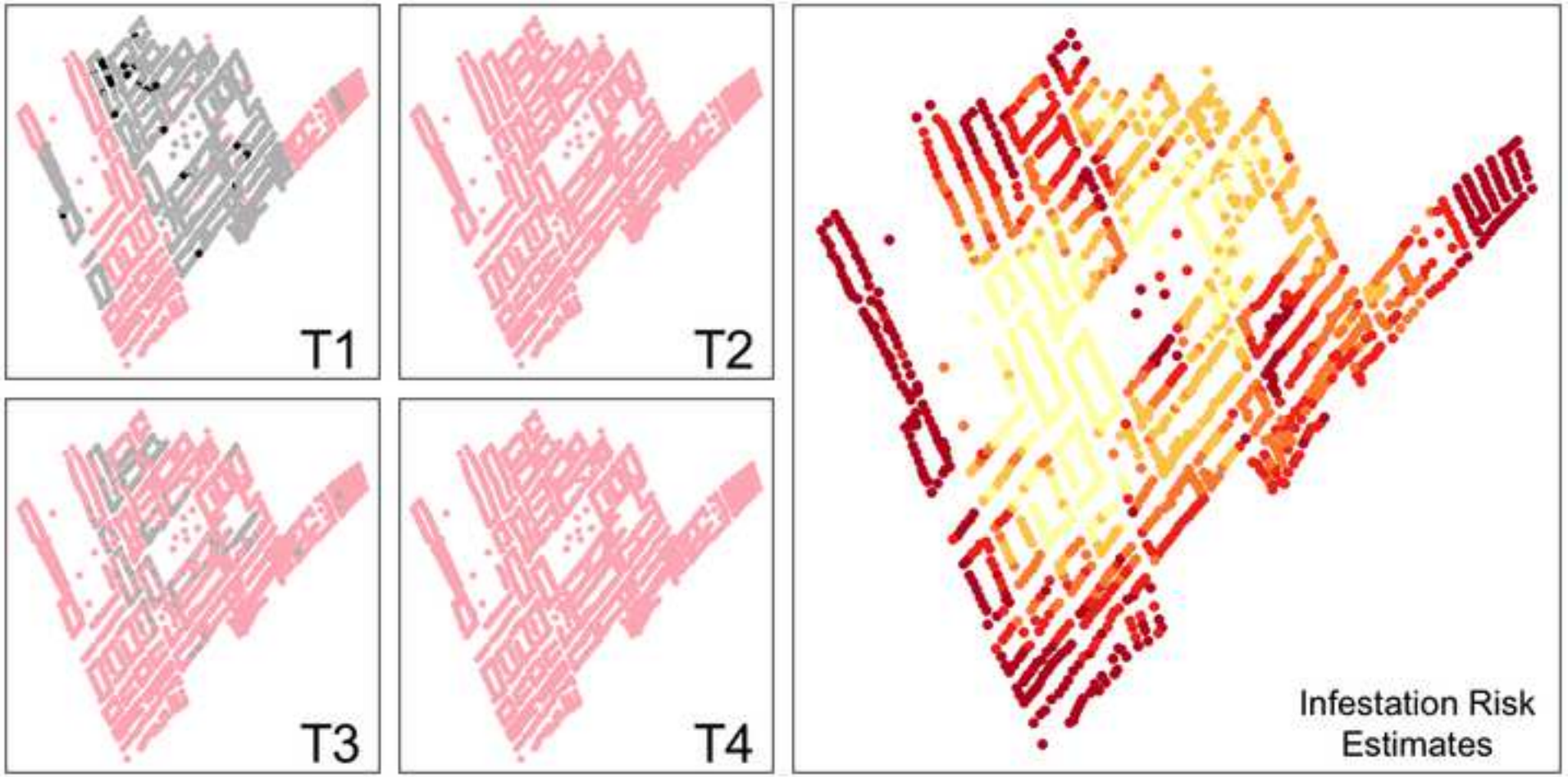


Figure 4

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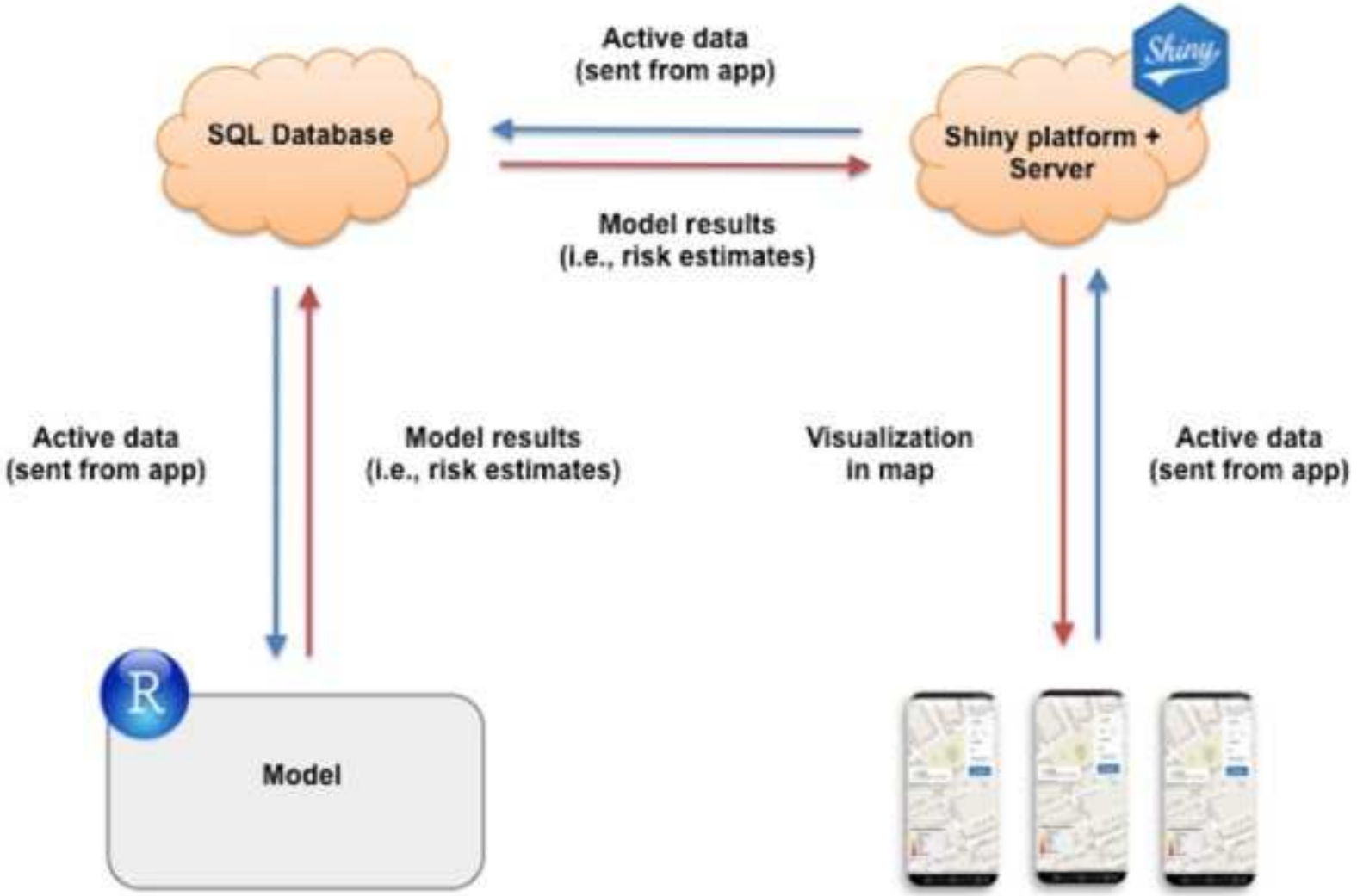


Figure 5

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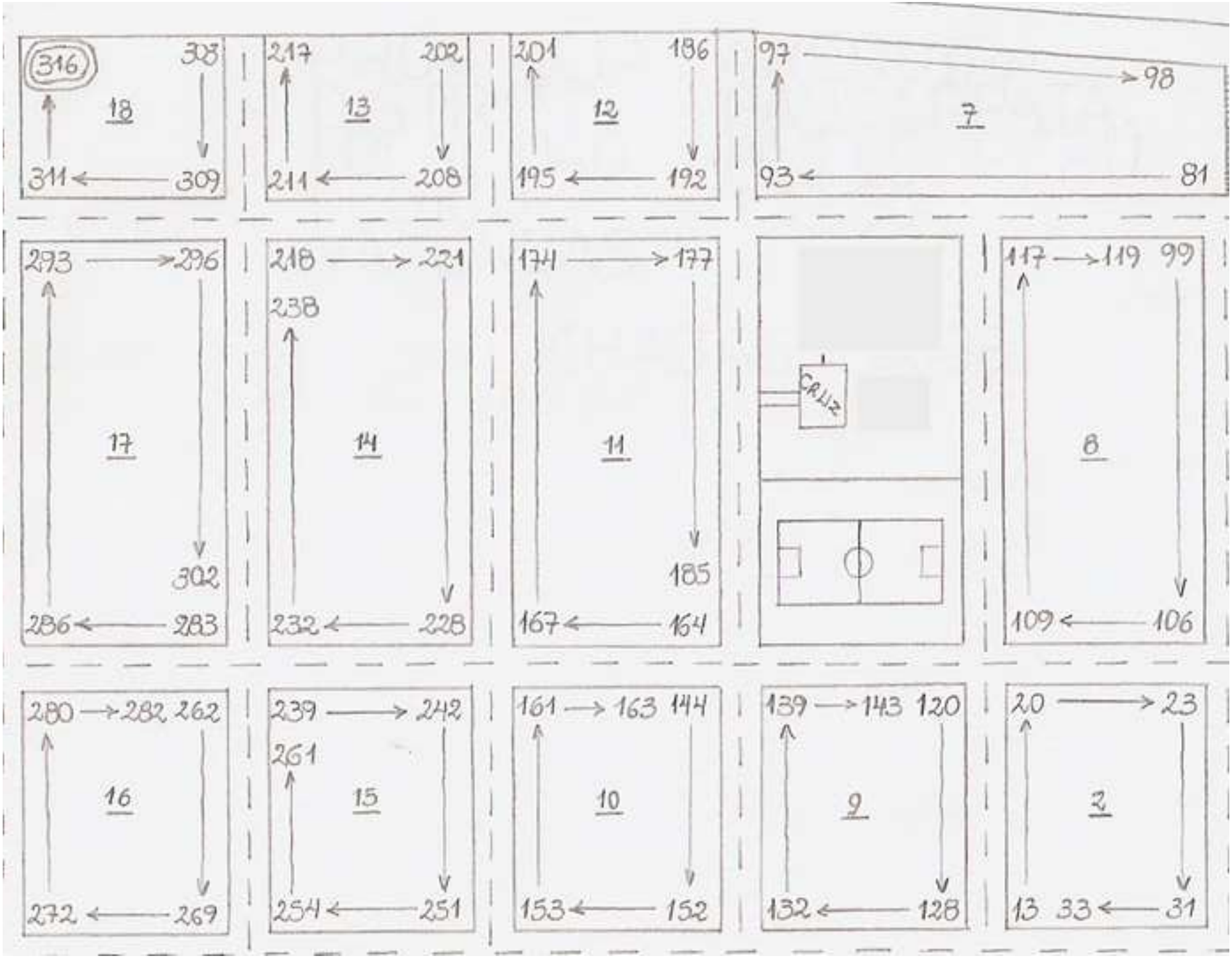


Figure 6

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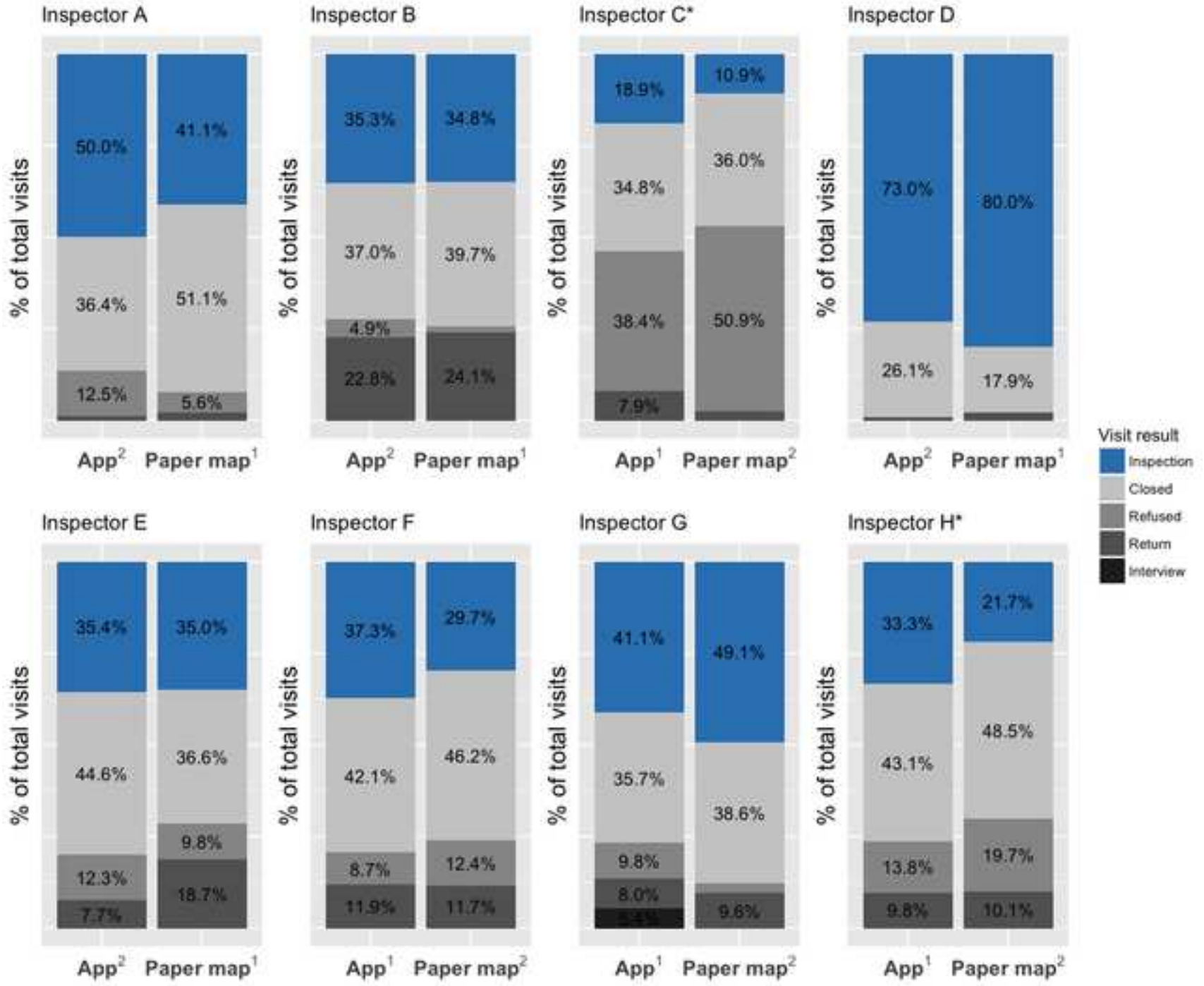
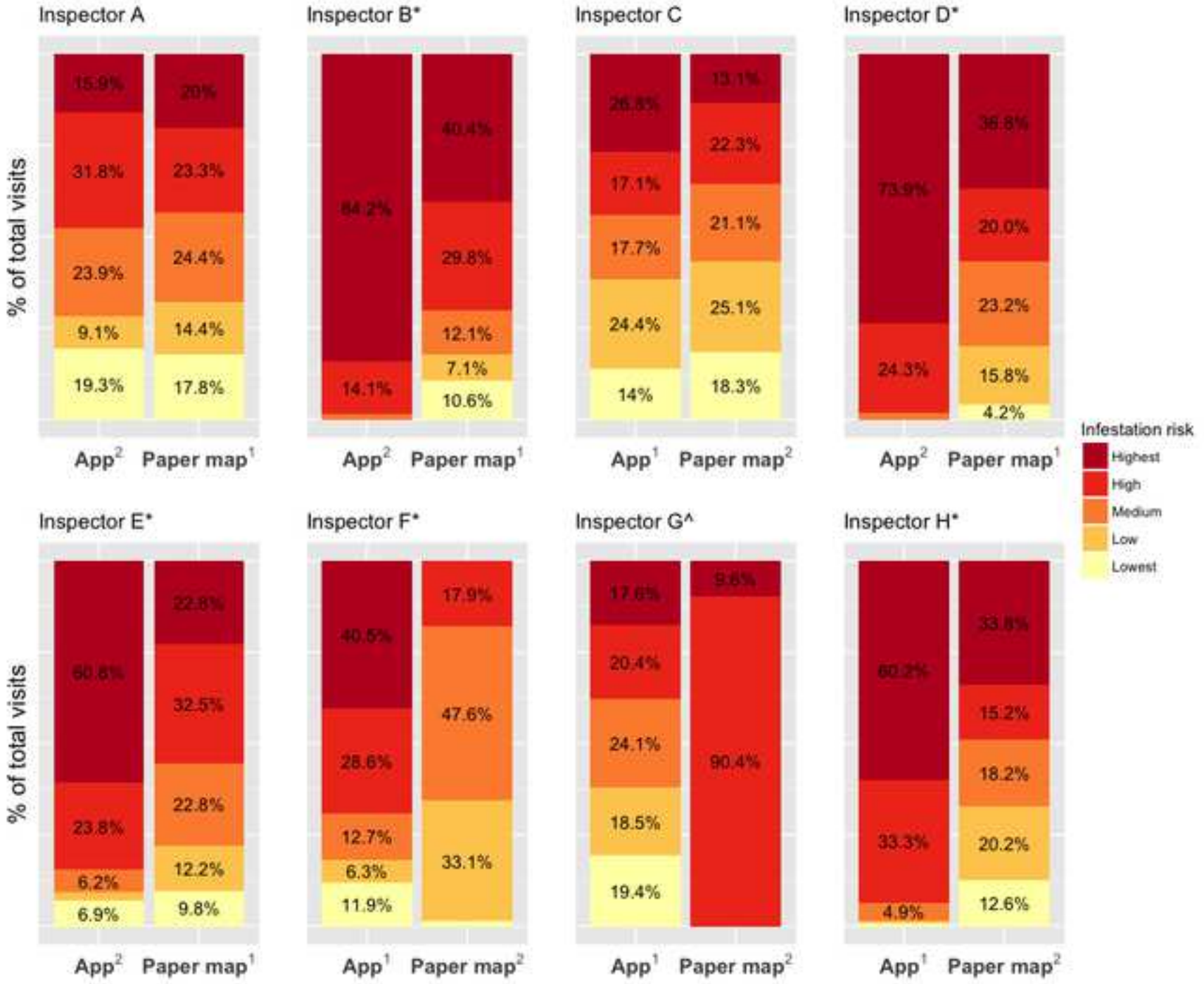


Figure 7

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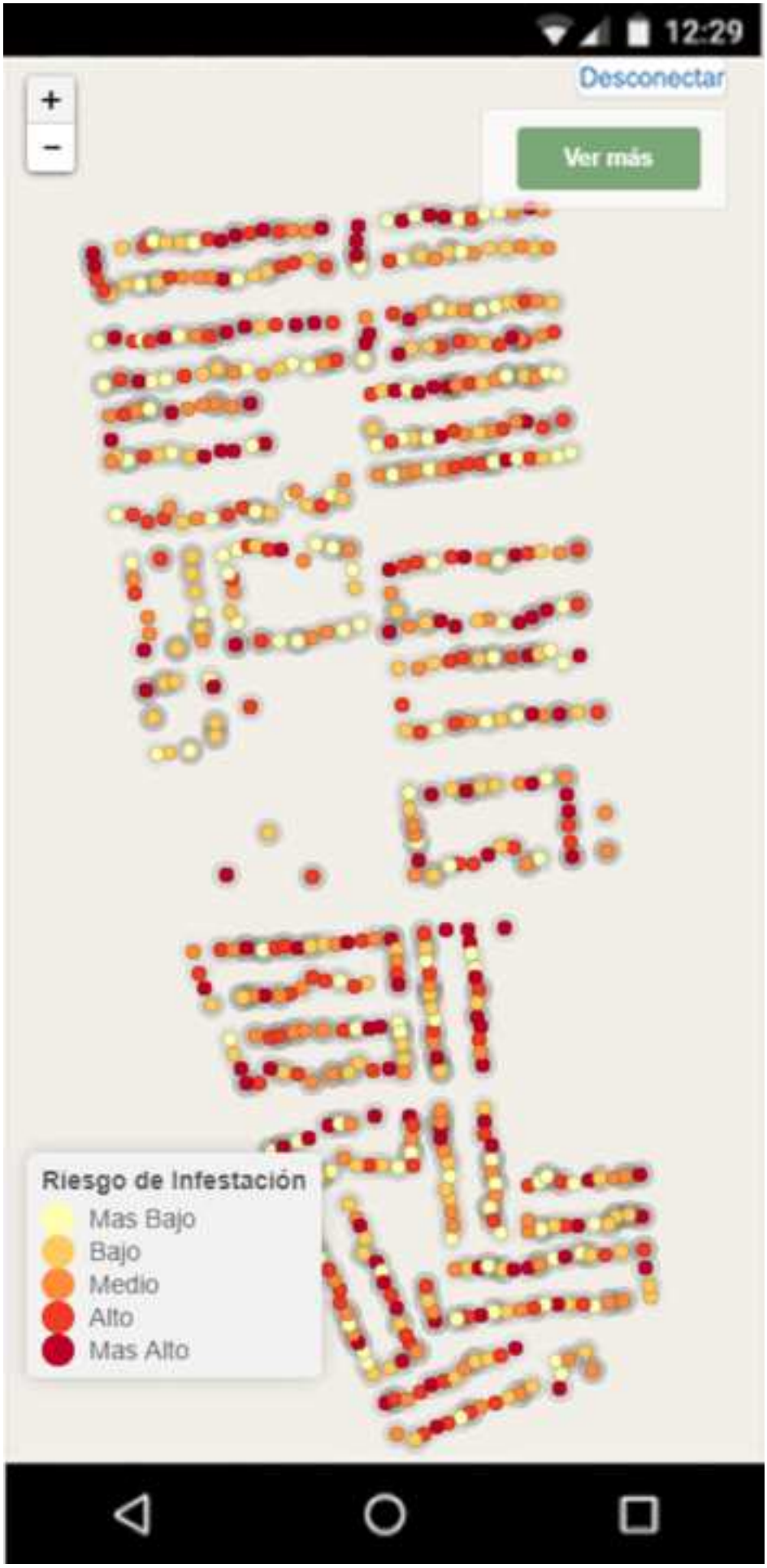


Pilot study data

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Supporting Information
S1_Dataset.xlsx

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September 19th, 2018

Dear Editors,

Thank you for the opportunity to revise our manuscript, ""Integrating evidence, models and maps to enhance Chagas disease vector surveillance" to be considered for publication as a Research Article in the PloS NTDs Vectors collection. The reviewers' comments are appreciated, and have helped us improve our manuscript. Listed below are point-by-point responses to each reviewer comment.

Sincerely,
Jennifer K. Peterson, PhD, on behalf of all authors

Responses to reviewer comments on the second submission

Reviewer #1: The study tackled two main quantitative questions: (i) 'app feasibility' and (ii) 'risk information use'. I strongly suggest that both be analyzed using GLMMs. I did this with the 'feasibility' data in my previous review, and the authors now have added a GLMM-based analysis of 'risk information use'. I think that both are quite informative and, therefore, that both should be presented in the manuscript. And I suggest that the key results that should be presented are (i) what exactly were the dependent and predictor variables; (ii) parameter estimates and their SEs/CIs (intercepts; fixed-effects betas/ORs; random-effects SDs); (iii) performance metrics (BIC scores probably better for the present analyses); and (iv) whether the models 'improved' by adding the app fixed effect (with 'improvement' measured by smaller BIC scores and smaller random-effects SDs). These GLMM-based findings should in my opinion be given more emphasis (perhaps in a Table) and be more thoroughly discussed.

== *Response* ==

We indeed ran the GLMMs for both feasibility and risk information use and appreciate the opportunity to report them fully. A paragraph with the GLMM results for app feasibility has been added to the manuscript, and we have also added in a table, as suggested (below). In addition, we include the following text in the methods and discussion sections to discuss the GLMMs more thoroughly:

"In the generalized linear mixed models (GLMMs), the odds ratio of the app fixed effect was 1.18 (95% CI = 0.97-1.42), suggesting little effect on the odds of a visited house being inspected. The standard deviation in inspector random effects was high regardless of whether or not the model included an app effect (SD = 0.76, 95% CI=0.49-1.39 with the app; SD = 0.77, 95% CI = 0.49-1.40 without the app). The addition of the app fixed effect did not improve the model, with similar BIC scores for both models (138.71 and 138.80 for the model with and without the app, respectively). Results from the GLMMs are presented in Table 2."

"In the GLMM, the app effect was a significant predictor of a higher risk house being visited ($p < 2e-16$), and it increased the odds of visiting a higher risk houses by 2.73 (95% CI = 2.24-3.32, Table 2). Standard deviation in inspector ID random effects was again high in both models (0.76, 95%

CI=0.48-1.37 with the app and 0.77, 95% CI = 0.50-1.20 without the app). BIC scores suggested that the app improved the model, with BIC scores of 442.9 and 544.3 for the models with and without the app, respectively. "

"we observed a significant upward shift among all inspectors in the number of high risk houses visited when using the app compared to paper maps, and a majority of the inspectors visited significantly more higher risk houses when using the app. These results were then confirmed in a second analysis comparing generalized linear mixed models run with and without the app, in which the the app was a highly significant predictor of a higher risk house being visited."

"While our results are encouraging, one inspector did not show any signs of using the risk information, and we cannot assume adoption would be universal. Indeed, our analyses using generalized linear mixed models confirmed that random among-inspector variation is high regardless of the outcome tested."

	Inspector productivity		**Risk information use**	
	Random inspector variation only	***With app fixed effect***	***Random inspector variation only***	***With app fixed effect***
Dependent variable	# inspections		# higher risk houses visited	
Predictor variable	Inspector variation	App use	Inspector variation	App use
Inspector variation intercept (95% CI)	-0.495 (-1.107-0.118)	-0.576 (-1.191-0.041)	0.571 (-0.031-1.177)	0.110 (-0.512-0.733)
Intercept SE	0.275	0.279	0.272	0.281
Intercept p-value	0.0723	0.0387	0.0356	0.697
App effect parameter estimate (95% CI)	n/a	0.163 (-0.026-0.352)	n/a	1.00 (0.808-1.202)
App effect parameter estimate SE	n/a	0.096	n/a	0.100
App effect p-value	n/a	0.0906	n/a	<2e-16
Estimated mean probability of outcome occurring (likelihood profile	0.379 (0.249-0.530)	0.398 (0.228- 0.597)	0.639 (0.492-0.765)	0.753 (0.573-0.874)

95% CI)				
Random effects SD estimate (95% CI)	0.767 (0.489-1.395)	0.764 (0.487- 1.391)	0.756 (0.484-1.374)	0.774 (0.495-1.202)
Odds ratio (95% CI)	n/a	1.18 (0.974-1.142)	n/a	2.73 (2.240-3.321)
BIC score	138.80	138.71	544.30	442.90
Model improved with app fixed effect?	no		yes	

Table 2. Results for generalized linear mixed models investigating inspector productivity and risk information use with and without the Vectorpoint app as a fixed effect.

And some small suggestions.
== Overall Response (specific responses below, where appropriate) ==
We thank the reviewer for the diligent comments, which were incorporated into the text in their entirety.

l. 51: delete comma before "(five with each..."
l. 54: check comma after "p < 0.001)"
l. 56: I suggest "odds ratio = 2.42, 95% CI 2.00-2.92)"
l. 77: consider "health personnel" instead of just "they"
l. 83: "...using THE app..."
l. 92: "...nine million..."
l. 94: "...or cure [4,5]. The etiological..."
l. 95: consider "parasite of mammals" instead of "mammalian parasite"
l. 95-6: "...by triatomine bugs..."
l. 108: "...infestation by the triatomine..."
l. 109: delete comma after "T. infestans"
l. 151: delete comma after "decision-making"
l. 157: "...carried out a pilot trial [right?] comparing..."
l. 210: I'd suggest that an explanation of what the finding of 'rastros' means be given here
l. 231-2 and 233-4: some repetition here ("indicator of participation in the attack phase")
l. 244: I think a brief explanation of the Matérn covariance structure, perhaps with a ref, would be helpful here
== Response ==
We have added a brief explanation and the following references:

1. Rasmussen CE, Williams CKI. Gaussian Processes for Machine Learning. MIT Press; 2006.
2. Lindgren, F., Rue, H., Lindstr¨om, J., 2011. An explicit link between gaussian fields and gaussian markov random fields: the stochastic partial differential

equation approach. Journal of the Royal Statistical Society: Series B (Statistical Methodology) 73 (4), 423–498.
3. Song H-R, Fuentes M, Ghosh S. A comparative study of Gaussian geostatistical models and Gaussian Markov random field models1. J Multivar Anal. 2008;99: 1681–1697.

l. 264: "...infested with Triatoma infestans..."
l. 274: what is the definition of "geographic median"?
== Response ==
We thank the reviewer for bringing this to our attention, as it should be 'mean' and not 'median,' because it refers to the center of a given geographic area (statistical mean of the coordinates), in this case it is the center of the block. We have changed this to read 'geographic center' and added in the definition.

l. 277-9: what was the SD of this prior on the intercept?
== Response ==
INLA requires that we set the precision matrix, which we set to 50, making the variance 1/50, and the SD 0.14. We have added this to the manuscript.

l. 312, 313, 314, 322, 537-9...: please check concordance (e.g., I think it should read "he or she enters", "the inspector selects [...] localities where she or he will carry out...")
== Response ==
The journal prefers the use of the gender neutral singular-they when referring to an individual (rather than 'he or she'), which is why the concordance may seem off in these cases.

l. 349: italicize T. infestans
l. 358: "..a paper map..."
l. 372: "...and feces. All inspectors..." (and I'm not sure I understand this bit: "inspectors were consented with a written informed consent")
l. 377: delete comma after "...of the house)"
l. 424: please explain briefly how exactly did you compare the results from those two models
== Response ==
We present both models with the BIC as a metric to compare them. For the productivity model the BICs are essentially equal, as is the interpretation--the app had little to no effect on the number of houses inspected. For the risk use model the BICs are very different, with much more support on the model that includes the app covariate. We have added this information to the revised manuscript:
"We used the BIC scores [87] as a metric to assess if the addition of the app fixed effect improved the performance of the models, with smaller BIC scores suggesting model improvement."

"The addition of the app fixed effect did not improve the model, with similar BIC scores for both models (138.71 and 138.80 for the model with and without the app, respectively)."

“BIC scores suggested that the app improved the model, with BIC scores of 442.9 and 544.3 for the models with and without the app, respectively.”

Reference:

87. Schwarz G. Estimating the Dimension of a Model. Ann Stat. Institute of Mathematical Statistics; 1978;6: 461–464.

l. 437 and elsewhere (e.g., l. 551): I'm not sure "feasibility" is the best word here...
l. 440-1 (and 463-4 and perhaps elsewhere): you don't need to repeat "chi square" ("Mantel-Haenszel test, chi square = 2.63" is ok)
l. 441 and elsewhere: I suggest that you give the abbreviation for 'odds ratio' (OR) here and use it in the remaining of the text
l. 440-1: in general, I'd suggest that, if you really want to keep the results of null hypothesis testing, you consistently give first the test results (test name, test statistic [when appropriate], and p-value), then the OR with CI
l. 443 and elsewhere (e.g., l. 452-62): use inspectors codes consistently (either "inspector 'A'" or just "inspector A")
l. 445 and eslsewhere: check punctuation marks (commas, semicolons, etc.) in these strings of numerical results (in this line we have ";," and "3.76 p = 0.046")
l. 448-9: please explain here what asterisks mean
l. 461-2: how did you compute this OR? (it seems to me that your data are separated - with 100% of houses being high-risk in the paper-map arm)
== Response ==
The data for inspector G were separated, because, while using the paper maps, inspector G visited only higher risk houses (114 higher risk, and 0 lower risk). While using the app, inspector G visited 41 higher risk houses and 67 lower risk. Therefore, the OR was calculated the same as the others (using the fisher.test function in R, which calculates the odds of the first cell being 0), but the data were very skewed.

l. 468-9: please see my general comment above about presentation of GLMM results...
l. 540: check "in at"
l. 548: consider adding "in urban Arequipa" after "Triatoma infestans"
l. 555: add a comma after "[85]"
l. 575-88: some readers may feel a bit curious about whether the app also increased bug-finding probabilities... Even if they're low, and the data too sparse to use them in a formal analysis, it would be nice to know, for example, what infestation indices would have been reported using maps and using the app
== Response ==
One infested house was found while using the paper maps. We have added this information to the text:
“In addition, motivation to use the app may increase as more data are collected, allowing us to compare infestation indices between houses inspected with the app versus paper maps. As

mentioned above, this study was not scaled to formally compare infestation indices; only one infested house was found in the study, and the inspector was using a paper map at the time."

l. 627: delete 'of' in "...it uses of entirely..."

Reviewer #3: Revisions adequately address reviewers comments

Reviewer #4: I still disagree that the most important objective of the study is to analyze data on how the collectors used the risk information, when the collectors were given no guidance on how they should use the information to conduct their searches. But we may have to agree to disagree on this point. There is extensive analysis in the results sections 'risk information use' and 'qualitative characterization of movement patterns' and in my view this is overkill for such a small sample size.
== Response ==
We thank the reviewer for allowing for our perspective on this first study of the surveillance app. We decided to start by determining if the app would be adopted by the inspectors, and to see how they used it. Therefore, we did not want to direct or dictate how they chose to use the app, but we thoroughly trained them in using the app and all inspectors had prior experience searching houses for triatomines. We include the qualitative information, despite the small sample size, because we find it informative and worth further investigation.

In my view the more important output from this study is a predictive risk model which is in an easy to use format, but the manuscript still lacks strong enough rationale for the parameters chosen. The evidence previously cited was incorrect and this has been removed, however the authors have not gone far enough to justify their choices. This could be addressed by including an overview of the risk factors for infestation in the introduction (including all risk factors, not just those included in the model), and by adequately describing it in the methods. Reference 30 includes much more nuanced information about the risks of re-infestation, and spatial auto-correlation which may have been taken into account with this model but is difficult to tell what decisions have been made based on what is written here.
== Response ==
We fit spatial autocorrelation to the data using the longitudinal INLA model--this is a great improvement over our earlier work (Barbu et al. EID), which used only an area-level random effect. The full technical details of the INLA model, previously available only in thesis form, are now in press at Spatial and Spatio-temporal Epidemiology. We have updated the reference. As for covariates, the app is meant to be used city-wide, and we therefore limited ourselves to the only covariates consistently available on that scale (80,000 houses). These variables are: (1) participation in the attack phase (binary-yes/no), and (2) house infestation status during the attack phase and all subsequent inspections. These covariates are included in the model; nothing is left out. The finer scale information, although it would be great to have, was not collected systematically across the whole city.

First Submission Responses

Reviewer Responses- comments to be addressed are in bold; author responses in italics

Methods

Reviewer #3: The objectives of the study are clearly articulated with an appropriate study design to address the objectives. The study population is clearly defined, and the sample size sufficient for statistical power.

Reviewer #4: **Regarding the methods to develop the VectorPoint model...Reference 23 is used to justify the use of previous surveillance visits as a risk factor, however this publication includes other highly significant risk factors (and insecticide use is not significant here although is one of the covariates in the model). It is not clear where this conclusion came from, and in general I suggest the authors provide a very strong rationale for the parameters chosen in the risk model.**

Author response: *This was a referencing error. Ref 23 is about risk factors for infestation on a very fine scale when detailed information is available from within houses. The models behind the app are based on Barbu et al. (Ref 30), which is limited to information available from the routine activities of the spray campaign, covering tens of thousands of houses. The very important risk factors such as guinea pig husbandry, just aren't available on the scale needed for the app. We have corrected the reference and clarified the reasoning for not using the information from Ref 23 in the text:*
*"*We did not include other finer-scale risk factors for *T. infestans* infestation, such as guinea pig husbandry [23], because data are not available at the scale required for app ."

The primary goal of the study is to determine whether collectors would use the risk information provided in the app to choose study sites. In my opinions the study design has significant limitations. First, **there was no training on how the collector should use the information and it appears that all collectors used the information although they may have had different goals – either to achieve equal coverage from all classes or to maximize the number of high risk households visited. Without clearly defined goals of the surveillance programme, it is flawed to test whether the collectors will perform as the study team secretly hopes they will.**

Author response: *We admit we did have secret hopes about how we wished the inspectors would use the app! The only surveillance goal stated was to find bugs (from the methods section: "*Inspectors were told to carry out *T. infestans* surveillance as they would normally, and that their objective was simply to find *T. infestans"). We consciously decided not to attempt to influence the inspectors in how they used the app in this first study, because we don't have any reason to believe that our preconceived ideas about how to use the app are better than what might emerge during the study--a marriage of the app and the intuition and experience of the field team.*

My second concern is that the data are presented as a trial but it is not clear what differences the trial is powered to detect. Given the very small sample size, this seems more like a pilot study where the findings should be used to further scale up and evaluate this tool in a larger trial.

Author response: *Yes this was a pilot study that we are hoping, (if we can convince funders that it is worthwhile), to scale up to a major trial, which we are currently designing. The trial will last 4 years and will be quite large, as the outcome we would test for is an increase in the rate of detection of triatomine infestations, which are (currently) rare. We have changed the wording in the manuscript from 'trial' to 'pilot study' to clarify. With any luck we'll be able to send a future manuscript on the trial results five years from now. We did not want to withhold the results from the pilot until then, including the code and corresponding software that goes into the app.*

In terms of primary outcomes of the study, the authors wanted to test whether the number of households visited and inspected differed between the two approaches. I suspect there are other variables which should be considered, including the number of visits that didn't result in inspection and the distance covered. For example, a greater number of total visits would be due to a lower number of inspections and vice versa**. The conclusion that the app did not hinder productivity is reasonable; however, the suggestion that it led to a greater proportion of houses being inspected for 2 inspectors is unfounded. Why would the proportion of houses that consented to inspection be dependent on the app use?**

Author response: *We agree, and have eliminated the sentence suggesting that the greater proportion of houses inspected by the two inspectors suggested an increase in productivity. In terms of the other variables to consider, distance covered is something we are very interested in, and we are working on a way to determine the exact path taken by each inspector. Our IRB was not comfortable with allowing us to use GPS tracking with the APP, so we were limited to only the location of each house visited, without the path taken between houses. For this reason, we included traveling in our qualitative analysis rather than quantitative analysis.*

It would have been very valuable to include qualitative methods to evaluate the users experience with both approaches, rather than just examining their paths covered. Information from a focus group discussion would provide strong justification for either further adapting the app or for scaling up the use.

Author response: *An excellent idea! We will plan focus groups for our future studies on the app.*

Reviewer #5: The Methods are appropriate and the relatively small number of participants sufficient to report on results attributed to the new technology. One minor query regarding the statistics is outlined in the attachment.

Results

Reviewer #1: Please see 'general comments' below

Reviewer #3: The analysis presented matches the analysis plan set out in the methods. The results are clear and well presented.

Reviewer #4: **What other data were captured in the app, is it possible to include any other analyses on the data capture process?**

Author response: *The primary data captured with the app in this pilot study were data from the entry forms used during inspections, and these forms are the same between the app and the paper maps. However, there are definitely other data capture capabilities we hope to integrate into the app's functionalities in the near future. At the top of our list, if the IRB allows it, would be high-frequency GPS measurements to determine the paths taken by each inspector. We can also gather information on animals inside houses--these data are not immediately helpful as covariates, as we can only gather them from houses that are entered, but over time they may prove helpful in the risk modeling.*

Were any positive households found, if so what was their risk rating?

Author response: *We did not find any positive households in the study. We have found positive houses using the app in the earlier searches during the app's development, but these were not done with any controls. As we mention above, it will take a large multiyear study to thoroughly evaluate the app and risk model behind it in terms of finding infested houses.*

Were there any patterns based on which approach was used first, for example did the individuals that used the app first perform differently than those that used the paper maps first?

Author response: *There was an even split. Of the five inspectors who visited more high risk houses with the app, two used the app first and three used the paper maps first. Of the two inspectors that did not visit more high risk houses with the app, one used the app first and the other used the paper maps first. We have added this information to the text.*
"We did not observe any patterns associated with arm order (i.e, which medium was used in the first week and which in the second week, Figure 7). Out of the five inspectors who visited proportionally more higher risk houses using the app, two used the app first and three used the paper maps first. For the two inspectors who showed no difference between the app and paper maps, one inspector started with the app, and the other started with the paper map. The inspector who visited more higher risk houses with paper maps used the app first."

Reviewer #5: Yes to all of the above, but see my comments.

Conclusions

Reviewer #1: Please see 'general comments' below

Reviewer #3: The conclusions are supported by the data presented. **The limitations may need to be reviewed to ensure they are highlighted.**

Author response: *We have added in the following text to the discussion and conclusion to highlight the limitations (new text is underlined):*
Discussion:
"However, it should be mentioned that although the VectorPoint app is flexible, the predictive model must still be adapted to fit the local epidemiological and ecological characteristics of each disease surveillance scenario to which it is applied, which could present challenges in cases where expertise is limited."

Conclusion:
"The app was designed to be used for *T. infestans* surveillance, but its components are cloud-based, open-source, and ready to be adapted to other appropriate scenarios, although the availability of sufficient data and/or resources may be a hurdle to overcome in some cases.

Reviewer #4: Overall this manuscript presents information on the development of the tool and describes how it can be used to improve data collection and decision making; **however, the study design makes it difficult to draw conclusions on whether this tool leads to improved surveillance.**

Author response: *We agree. We are in the process of designing a much larger study to assess whether the app will really improve the rate of detection of triatomines. Given the low level of infestation, the study being designed is very large (it's powered at 60 clusters of 3,000 houses each over a follow-up of 4 years). However, we believe it is important to disseminate the results from this study while we await the results from the large trial--especially the code for the app. Whether the houses are truly at higher risk is also a fair point. The risk model comes from retrospective analyses; as more data are collected using the app, the risk parameters can be adjusted. More importantly, as the app is applied to other datasets, it can be adapted to other risk factors (and other levels of data completeness).*

Reviewer #5: The authors come to the appropriate conclusions based on the data, and discuss them in the broader context. See my suggestions for inclusions re discussing the application of the technology

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Editorial and Data Presentation Modifications?
Reviewer #1: Please see 'general comments' below
Reviewer #3: (No Response)
Reviewer #4: **Abstract line 31 should say "thanks"**
Author response: *We thank the reviewer for pointing this out. Thanks has been replaced by 'attributed to.'*

Reviewer #5: minor revisions

Summary and General Comments
Reviewer #1: This is an interesting and well-written manuscript describing and testing a novel tool for triatomine bug surveillance – a mobile app that can be used both for entering field-survey data and, importantly, for quickly assessing model-based estimates of the relative risk of house infestation by the main local vectors (Triatoma infestans in Arequipa, Peru, where the project was developed). The sophisticated predictive models make use of large amounts of data, including rich historical information and virtually real-time input, to produce interactive maps in which each house in the neighborhood field workers are visiting is given a color-coded risk score. This, in principle, gives field inspectors the opportunity of selecting houses that the model predicts are at higher risk of bug infestation – which was not possible with the traditional, paper maps they used so far. On the other hand, the introduction of a new technology may add some difficulty to routine fieldwork – which might perhaps show up as lower productivity in terms, e.g., of how many houses are visited and inspected for bugs per person and unit time. With these possibilities in mind, the authors designed a trial to see whether the app increased the probability that higher-risk houses were visited (i.e., whether inspectors made effective use of the wealth of information the app provides) and whether there was any sign that the app was slowing fieldwork down. For this, eight inspectors used either the app or paper maps in their routine duties for one week, and then switched tools for another week.

I think this report will be a very valuable addition to the literature, and that the approach taken by the authors may help to strengthen vector surveillance – although it should perhaps be noted that the development and implementation of the sophisticated predictive models they used requires statistical skills (and data) that may not be readily available in most real-life routine surveillance settings. Along the same lines, **it is also important to highlight that such models must be adapted to local circumstances – vector species, housing characteristics, control tactics, and so on. Thus, I'd suggest that the authors stress these potential hurdles to the wider applicability of their app-based approach.**

Author response: *We agree completely, and have added the following text to the manuscript:*
"...it should be mentioned that although the VectorPoint software is flexible, the predictive model must still be adapted to fit the local epidemiological and ecological characteristics of each vector or disease surveillance scenario to which it is applied. It also requires some familiarity with spatial modeling in R, which will present challenges in cases where such expertise is limited."

As a second general point, I found it a bit odd that while the authors used highly sophisticated models for risk prediction, they chose a rather crude analytical strategy (described in lines 386-395) to assess the app's performance. **I'd rather recommend that such a null hypothesis-testing (NHST) exercise be presented as a preliminary (exploratory) step – which, in addition, I'd probably base on 95% CIs rather than on p-values. A second, inferential step should then follow in which some important details are taken into account.** The most important of those details is probably that assuming that each inspector's decisions and actions are independent is, in my view, hard to justify – and there seems to be very little doubt that the authors have the knowledge and skills needed to overcome such issues.

For example, I reanalyzed the "visit - inspection" data in Table I using binomial GLMMs with "Inspector ID" random intercepts and testing whether the inclusion of an "App" fixed effect would improve model performance (as measured by BIC and AICc). I found that:

a) The simpler model (glmer(Z~1+(1|id), family=binomial, data=data)) was barely improved by adding the App effect;

b) The mean estimated probability of inspecting a visited house was 0.3787 (likelihood profile 95%CI 0.2485-0.5296), irrespective of app/map use (simpler model); this is slightly larger than the sample proportion (0.3620), and, importantly, has a CI that is much wider than Fisher's exact CI (0.3447-0.3795);

c) The estimated among-inspector (random effects) SD was 0.767 (95% CI 0.489-1.395) (simpler model above);

d) The "App" fixed-effect model (glmer(Z~app+(1|id), family=binomial, data=data)) had a BIC score 4.80 units larger than the simpler model; the AICc score, however, was slightly smaller (0.86 units) for the App model (which reflects the fact that AIC tends to select more complex models that are better at prediction and BIC simpler ones that better approximate the 'true' model; see Aho et al. Ecology 2014; 95:631-6);

e) Finally, the estimated App odds ratio in the second model was 1.177 (95%CI 0.974-1.422), so that this appears to be what TW Arnold (J Wildl Management 2010;74:1175-8) called an "uninformative parameter"... The random-effects SD estimate was only slightly smaller (0.764) than that in the simpler model, also suggesting that the App fixed effect barely helped explain any variance.

From this I infer that there was no discernible effect of the App on the probability that an inspector inspected a house (s)he visited, but that there was a good deal of among-inspector variation in such a probability... When this variability is taken into account (simpler model, SD = 0.767), one may expect that approximately 95% of inspectors inspect approximately between 12% and 73% of the houses they visit (in the sample, inspector C inspected 15% and inspector D 76%).

I understand that this is very similar to what the manuscript reports, but **I suggest that it is a more rigorous approach (e.g., because it addresses inspector-level pseudoreplication) – and a more transparent one because it makes uncertainty plainly explicit in the form of (wide) CIs. I hence suggest that this example of reanalysis can, and possibly should, be used as a template for analyzing the rest of the app performance metrics** – choosing, of course, the appropriate error distribution families and link functions. In general, this suggestion

includes that p-values from NHST be avoided and replaced with point estimates and associated uncertainties (SEs and/or CIs) for the relevant parameters.

Author response: *We agree that a second, more rigorous analysis of app performance is useful, and we've carried out as requested. Odds ratios and corresponding 95% CIs have been added to the preliminary (i.e.,simpler) analysis results, and GLMMs have been run as a second, more rigorous analysis. We have added the following text to the methods, results and discussion reflecting the changes made to the current (i.e., simpler) analyses, and new results from the additional analyses:*

Methods:

"...we carried out a preliminary analysis using Fisher's exact tests to test for differences between the app and paper maps for each individual inspector, and a Mantel-Haenszel chi-squared test with continuity correction to test for an overall shift in one direction among all inspectors. We carried out a second analysis using binomial generalized linear mixed models with "Inspector ID" random intercepts to test whether the inclusion of an "App" fixed effect helped to explain any variance. In this analysis, we compared the results from a model run with only inspector ID random intercepts with results from a model run with inspector ID random intercepts plus an app fixed effect. Models were fit by maximum likelihood (Laplace Approximation) using the 'glmr' function in the lme4 package [84] for R. "

References

84. Bates D, Mächler M, Bolker B, Walker S. Fitting Linear Mixed-Effects Models Using lme4. Journal of Statistical Software, Articles. 2015;67: 1–48.

Results:

"In the generalized linear mixed models (GLMMs), the odds ratio of the app fixed effect was 1.18 (95% CI = 0.97-1.42), suggesting little effect on the odds of a visited house being inspected. The standard deviation in inspector random effects was high regardless of whether or not the model included an app effect (SD = 0.76, 95% CI=0.49-1.39 with the app; SD = 0.77, 95% CI = 0.49-1.40 without the app)."

"In the generalized linear mixed model the app effect increased the odds of a visiting higher risk houses by 2.73 (95% CI 2.24-3.32). Standard deviation in inspector ID random effects was again high in both models (0.76, 95% CI = 0.48-1.37 with the app and 0.77, 95% CI = 0.50-1.20 without the app).

Discussion:

(new text is underlined)

"...we observed a significant upward shift among all inspectors in the number of high risk houses visited when using the app compared to paper maps, and a majority of the inspectors visited significantly more higher risk houses when using the app, which we confirmed in a second, more rigorous analysis."

"While our results are encouraging, one inspector did not show any signs of using the risk information, and we cannot assume adoption would be universal. Indeed, more rigorous analyses using generalized linear mixed models confirmed that random among-inspector variation is high regardless of the outcome tested."

I finally note that there are some typos in the manuscript. For example, the very first sentence of the Abstract has three [suggestions in square brackets]: "Until recently, the Chagas disease vector Triatoma infestans, [delete comma or add a comma after 'vector'] was widespread in Arequipa, Peru, but hanks [thanks] to a decades-long campaign in which over 70,000 houses [were] treated with insecticides, infestation prevalence is now drastically reduced." And in the second sentence: "…a surveillance phase where a sample of houses are [is] selected for inspection…" I suggest that the authors carefully check their text again to correct these small issues.
Author response: We thank the reviewer for pointing out these errors, and they have now been corrected.

Reviewer #3: This paper focuses on integrating evidence, models and maps to enhance Chagas disease vector surveillance. Overall, it is well written, **however I found the style of writing a little informal in the abstract and introduction section and could be better worded.** The study is described in detail and very thorough, and the app seems to be a useful novel tool for surveillance.
Author response: *We thank the reviewer for this suggestion. We have replaced less formal words in the abstract and introduction with more formal ones:*
-'drastically' replaced by 'greatly'
-'thanks to' replaced by 'attributed to'
-'hugely difficult' replaced by 'highly challenging'
-'enormous' replaced by 'great'
-'a handful' replaced by 'a small number'

Minor comments
Abstract second line – spelling 'thanks'
Author response: *We thank the reviewer for pointing this out. We have replaced 'thanks' with 'attributed to'*

Abstract third line – the word 'significantly' may be better than 'drastically'
Author response: *We thank the reviewer for this suggestion. To avoid any implication of statistical significance, we have changed the word to 'greatly.'*

Are there any issues with confidentiality – i.e. risk or stigma associated with identifying individual houses at risk?

Author response: *Yes it is safe to assume there is a stigma associated with having an infested house. In fact one of the motivations for developing the app was to increase privacy--believe it or not, the 'old' way of tagging houses for future surveillance was painting a small cross beside the door of infested houses during the attack phase. The app is for use only by inspectors working with the public health department, and not for the general public, all data sent between the app and the databases are encrypted and very little identifying information (household code, location) are collected. We have added a line in the methods section clarifying this point:*

*"*All connections are encrypted, and risk maps can be accessed only by the study team and authorized health personnel."

Comment in discussion on the human variability of wanting to use technology/the app e.g. inspector G showed no indication of using the risk information in the app – why was this so, were they older and not familiar with technology. The success of the app will be related to who uses it.

Author response: *Due to the small number of inspectors in the study, the inclusion of demographic information would have made it obvious to the other participants and researchers who each inspector was. We have added new text and a new citation addressing the variability in adopting new technologies (new text is underlined)*

" While our results are encouraging, one inspector did not show any signs of using the risk information, and we cannot assume adoption would be universal. Willingness to use the information in the app may be associated with factors specific to the individual such as age or experience with technology [85], and we may need to to take into account inspector demographic information when analyzing app use in future studies in order to develop strategies for increasing adoption. Specialized training or more creative solutions such as incentives [88-90] or games [91,92] may be necessary in cases where inspectors have trouble engaging with the app, or if enthusiasm for the new technology wanes over time."

References:

85. The MAPS Toolkit: mHealth Assessment and Planning for Scale. Geneva: World Health Organization; 2015.

88. Behrman JR, Gallardo-García J, Parker SW, Todd PE, Vélez-Grajales V. Are Conditional Cash Transfers Effective in Urban Areas? Evidence from Mexico. Educ Econ. 2012;20: 233–259.

89. Van Herck P, De Smedt D, Annemans L, Remmen R, Rosenthal MB, Sermeus W. Systematic review: Effects, design choices, and context of pay-for-performance in health care. BMC Health Serv Res. 2010;10: 247.

90. Emanuel EJ, Ubel PA, Kessler JB, Meyer G, Muller RW, Navathe AS, et al. Using Behavioral Economics to Design Physician Incentives That Deliver High-Value Care. Ann Intern Med. 2016;164: 114–119.

91. Garard DL, Lippert L, Hunt SK, Paynton ST. Alternatives to traditional instruction: Using games and simulations to increase student learning and motivation. Commun Res Rep. Routledge; 1998;15: 36–44.

92. Clark AM, Clark MTG. Pokémon Go and Research: Qualitative, Mixed Methods Research, and the Supercomplexity of Interventions. International Journal of Qualitative Methods. SAGE Publications Inc; 2016;15: 1609406916667765.

Reviewer #4: This manuscript presents the VectorPoint model which assigns an infestation risk to a household based on whether the household was previously surveyed. The statistical model was developed into an app which is also used for data collection. Undoubtedly the data collection function will be an improvement over paper-based records **but what is still unclear is whether this model is able to identify the true high-risk households, utilising so few variables, and whether targeted surveillance according to the model predictions is more likely to identify positive households than the current surveillance method. Without validating this model in some way, or providing strong enough rationale that whether a house was previously visited is the single most important indicator of infestation, it is hard to say which surveillance method would be most appropriate for programmatic use.**

Author response: *We agree. We are in the process of designing a much larger study to assess whether the app will really improve the rate of detection of triatomines. Given the low level of infestation, the study being designed is very large (its powered at 60 clusters of 3,000 houses each over a follow-up of 4.5 years). In the meantime we believe it is important to disseminate the research carried out using the app so far, to share its capabilities and to offer the open source code for its use, while we await results from the larger trial. Whether the houses are truly at higher risk is also a fair point. The risk model comes from retrospective analyses; as more data are collected using the app, the risk parameters can be adjusted, and, more importantly, as the app is used with other datasets, it can be adapted to other risk factors (and other levels of data completeness).*

Reviewer #5: Strengths- novelty and a topical study conducted by experts; a study that is largely observational without being over ambitious or claiming things that the data do not support.
Weakness - only with respect to expanding the discussion as suggested, though not a major concern.
Author response: *Please see our expansions to the discussion described below.*

Reviewer 5- comments from the attachment

The manuscript reports on a novel interactive mtool for surveillance against household infection with triatomine vectors. This is a topical subject, well written, with interesting observations on its effectiveness in changing health inspector search behaviours. There are no major issues with the manuscript, and in my opinion worthy of publication in PloS NTDs.

Discussion points

It would be informative if the authors could further place the use of the mtool into specific context of surveillance. This would help in the interpretation of the results beyond the primary aim to demonstrate that inspectors are using the additional risk information in VectorPoint to change their search patterns. For example, is the premise that visiting/inspecting high risk houses is more desirable or effective than visiting/inspecting low risk houses? Is this the primary objective for control /elimination, and has it been shown to maintain low transmission levels elsewhere? Related to this, **the authors could discuss in what way use of this additional risk information in changing search patterns will enhance surveillance or conversely might result in unwanted sampling bias under certain circumstances. E.g. is providing high risk house locations promoting "finding" high risk houses/clusters, rather than detecting new high risk houses/hotspots**?

Author response: *We think visiting higher risk houses is preferable to the traditional mode of surveillance, which was not well informed by historical infestation and surveillance data as well as frequently being haphazard or based on convenience. But we agree that what is really exciting is that the app will allow us, with a lot more work, to go into greater depth of optimal search patterns, which would combine the goal of visiting higher risk houses with the goal of improving our risk estimates (ie optimizing the 'exploration/exploitation tradeoff'). We're going step by step, and the first step we report here is that inspectors are, for the most part, adapting their surveillance routines to integrate the new information presented in the app, at least over the short term. We have added the following text to address the reviewer's final point about promoting the searching of houses or areas already predicted to be high risk vs. detecting new hotspots.*

"Extensions to the app, or the model itself, may be useful in improving spatial coverage of surveillance, and for exploring more complex questions revolving around the benefits or drawbacks of using the app. In particular, we do not know if a trade off occurs between exploration and exploitation when using the app; by focusing on areas that are known or predicted to have a higher risk of infestation (i.e., exploitation), will we fail to detect new high risk foci (i.e., exploration)?"

The trial represents the first use of the novel technology tool by the inspectors, so it can be assumed that their enthusiasm to use the risk information provided was high. The authors could comment on the possibility that the enthusiasm might wane with time e.g. by citing examples about user continuity for technology driven surveillance tools elsewhere.

Author response: *We thank the reviewers for bringing up a good point. We have added a note to the text that enthusiasm might wane over time, and suggest some possible means to inject novelty into the app to keep it fresh and exciting.*

*"*Specialized training or more creative solutions such as incentives [88-90] or games [91,92] may be necessary in cases where inspectors have trouble engaging with the app, or if enthusiasm for the new technology wanes over time."

References

88. Behrman JR, Gallardo-García J, Parker SW, Todd PE, Vélez-Grajales V. Are Conditional Cash Transfers Effective in Urban Areas? Evidence from Mexico. Educ Econ. 2012;20: 233–259.

89. Van Herck P, De Smedt D, Annemans L, Remmen R, Rosenthal MB, Sermeus W. Systematic review: Effects, design choices, and context of pay-for-performance in health care. BMC Health Serv Res. 2010;10: 247.

90. Emanuel EJ, Ubel PA, Kessler JB, Meyer G, Muller RW, Navathe AS, et al. Using Behavioral Economics to Design Physician Incentives That Deliver High-Value Care. Ann Intern Med. 2016;164: 114–119.

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92. Clark AM, Clark MTG. Pokémon Go and Research: Qualitative, Mixed Methods Research, and the Supercomplexity of Interventions. International Journal of Qualitative Methods. SAGE Publications Inc; 2016;15: 1609406916667765.

Specifics

Lines 196-8. "The maps are set to display relative risk (i.e., a house's risk of infestation relative to all other houses in the neighborhood), but they can be adjusted to display data divided into any number of quantiles, or to display absolute, instead of relative, risk estimates". Can the settings be altered by the inspectors, and if so, state if they had this option during the trial and how this might have varied their individual searching behaviour.

Author response: *Providing an option for the end user to display absolute risk would be fairly simple, although it may have unintended consequences--an inspector in a low risk area might lose motivation to search. It's something that needs further study, and so in this first pilot with the app, the inspectors did not have the option to change any settings for how risk was displayed. We have added text to the manuscript to clarify this:*

"[the maps] can be adjusted from the back end to display data divided into any number of quantiles, or to display absolute, instead of relative, risk estimates."

Lines 204-5. Describe the methods used in routine house inspection for triatomines – e.g. was it timed manual capture?

Author response: *Inspections consist of systematic searches in all areas of the domicile and peri-domicile (pending permission by the resident), including spaces occupied by humans and live animals. Searches last approximately one person-hour; flexibility is allowed to account for the heterogeneity in the size of houses. During the search, the inspector looks for live* T. infestans*, eggs, exuviae, and insect feces. We have added this text to the manuscript:*

“The protocol for *T. Infestans* inspections is stipulated by the Peruvian Ministry of Health, and consists of systematic searches in all areas of the domicile and peri-domicile (pending permission by the resident), including spaces occupied by humans and live animals. Searches last approximately one person-hour*;* flexibility is allowed to account for the heterogeneity in the size of houses. During the search, the inspector looks for live *T. infestans*, in addition to *T. infestans* eggs, exuviae, and feces.”

Line 272. Is 1 in 1000 based on empirical data? In general, state the assumptions of the model.

Author response: *The 1 in 1000 prior set in the model for the expected baseline infestation prevalence in the area is based loosely on the detection rate for active surveillance (i.e., house to house surveillance by health personnel) in 2017, which was 6/1000, with the sample considered to be houses inspected. Much of that active surveillance was targeted in areas near reported infestations (i.e., passive surveillance), and 6/1000 was the highest rate in years (active surveillance between 2009-2016 yielded a median of 2 in 1000). From passive surveillance (i.e., reports of* T. infestans *infestation from community members that are later confirmed by health personnel), confirmed reports are consistently less than 1 in 1000 when the sample is considered to be all houses under surveillance. In the end, to choose our prior, we used this conservative estimate. We have clarified this in the manuscript:*

"[1 in 1000] reflects our best estimate of *T. infestans* infestation prevalence in Arequipa based on recent results from both passive surveillance (i.e., reports of *T. infestans* infestation from community members that are later confirmed by health personnel) and active surveillance (house to house surveillance conducted by our team and the Ministry of Health)."

Line 386. Apart from inspector ID and trial arm, there are potential confounders e.g. city zone, topography, date, etc., that should be accounted for / used to adjust the effect estimates, in the statistical analyses.

Author response: *We did our best to choose search zones that were comparable. All search zones were in the same district in Arequipa, and contained similar topography, numbers of homes, area, house density (#homes/sq km) and infestation history. While we agree there are many potential confounders, we are hesitant to include, post hoc, additional variables that were not prespecified in the trial design, into a model. We have added the criteria for the search zones to the text:*

‘..all zones were in the same city district. Each search zone met the following five criteria: (i) it was located in a developed area (i.e., all roads paved) within the central portion of the district, (which is safer than peripheral, less developed areas); (ii) it contained a minimum of 400 houses, and no more than 1.25 times the number of houses in the zone with the fewest houses; (iii) area was a minimum of 0.1 km^2, and could not be greater than twice that of the zone with the smallest area; (iv) house density was at least 2000 houses per km^2; and (v) the search zone was in a locality where at least one house had been found positive for *T. infestans* during the attack phase of the control

campaign. These criteria resulted in 16 search zones with 416-514 homes, an area of 0.12-0.20 km^2, a perimeter of 1.62-2.37 km, and house density ranging from 2,570 to 3,623 houses per km^2."

Line 552. What is the spatial kernel in the model- how does adjusting the risk of one house influence the risks of neighbouring houses?

Author response: *We are using a a very close approximation to a Gaussian field in the model. It's like a spider web--any adjustment to the estimate for one house affects all the other houses, with a greater effect on those nearby. But, unlike a kernel, the parameters of the Gaussian field aren't particularly interpretable. We have added text to clarify (new text is underlined):*

*"*The back end of VectorPoint (Figure 2) is composed of a spatio-temporal Gaussian field model that generates the infestation risk estimates visualized in the maps…"

"We model the probability using a logistic model with intercept, β_0, covariate information, β_1, and separable spatio-temporal random effects, ω_{st} and ω_{st}:

$$\text{logit}(p_{st}) = \beta_0 + \beta_1 + \omega_{st} + \omega_{st}$$

where ω_{st} is a realization of the Gaussian field with a Matern covariance structure. <u>The Gaussian field functions such that any adjustment to the estimate for one house affects all the other houses, with a greater effect on those nearby.</u>

Figure 5. The legend should explain the arrows on the maps.

Author response: *We have added the following text to the legend:*

"The numbers displayed on each block are the last three digits of the house code, with arrows indicate their ascending order. For example, a block with '1 → 12' indicates that houses with unicodes one through 12 are on that block."

Figure 6. Comment on the clear difference of inspector D with respect to % houses inspected. Was it that he/she was more persuasive than the others to gain access- or maybe it is a more compliant city zone

Author response: *It seems to be an inspector effect (rather than a city zone). We have added the following text to the manuscript addressing inspector variation (new text underlined) and potential methods for increasing engagement:*

"While our results are encouraging, one inspector did not show any signs of using the risk information, and we cannot assume adoption would be universal. <u>Indeed, our more rigorous analyses using generalized linear mixed models confirmed that random among-inspector variation is high regardless of the outcome tested. Willingness to use the information in the app may be associated with factors specific to the individual, such as age or experience with technology [85], and we may need to take into account inspector demographic information when analyzing app use in future studies in order to develop strategies for increasing adoption. . Specialized training or</u>

more creative solutions such as incentives [88–90] or games [91,92] may be necessary in cases where inspectors have trouble engaging with the app, or if enthusiasm for the new technology wanes over time."

References:

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